

# Introduction to Geodesy

Basics, GNSS Techniques



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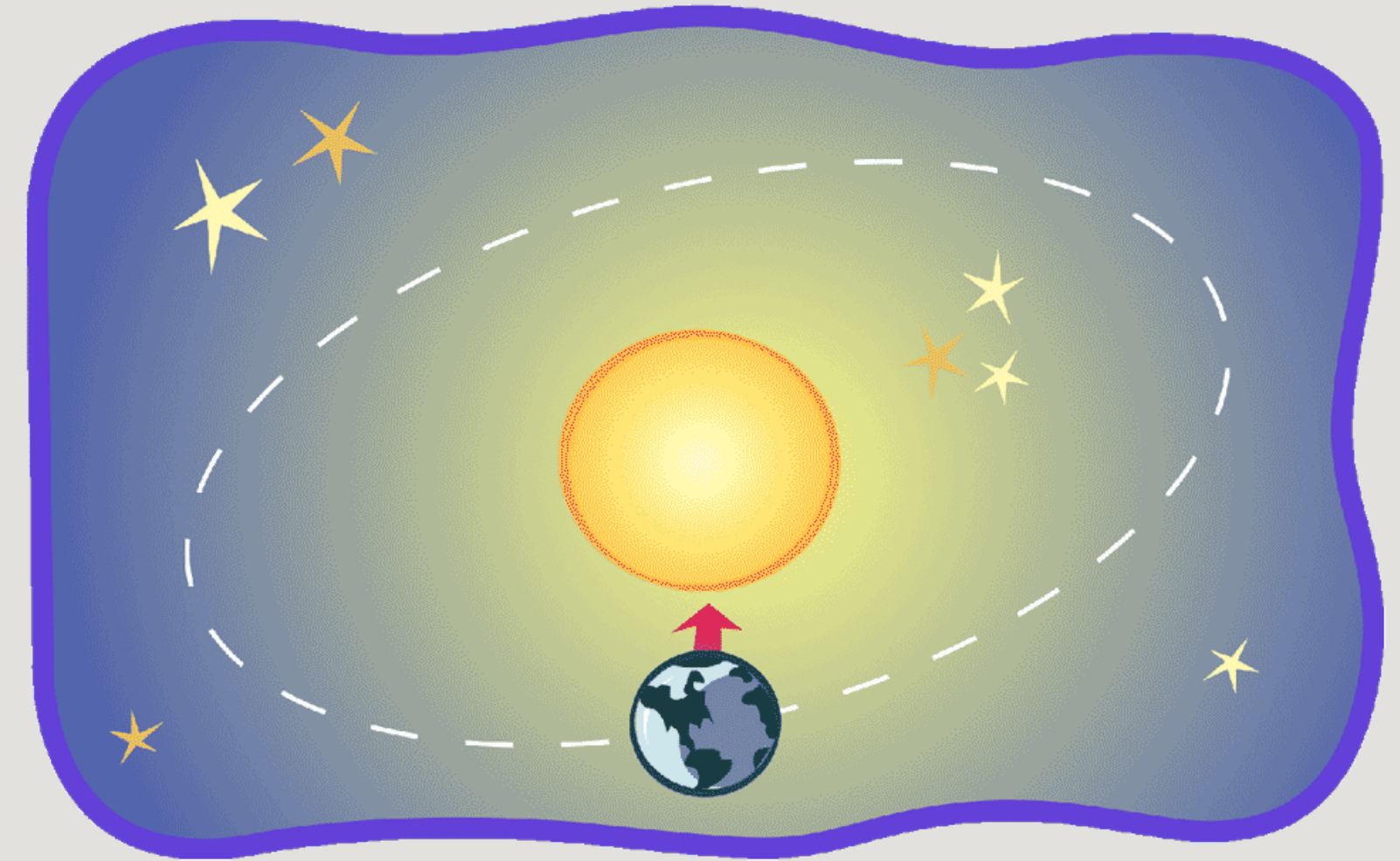
**GNSS Techniques**



# What is Geodesy?

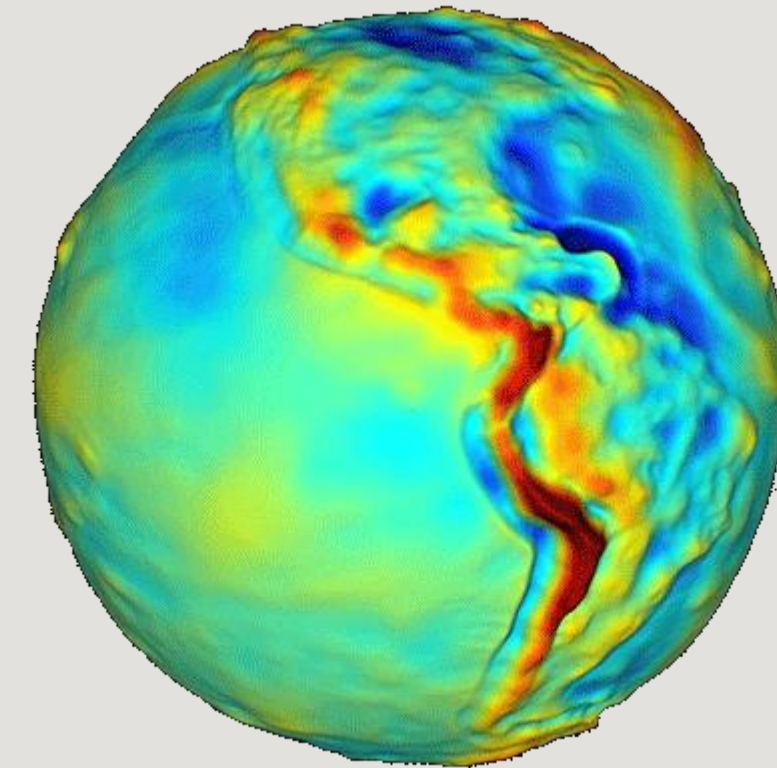
## Understanding the Shape, Size and Gravity of the Earth

- Geodesy is the science of measuring and understanding the Earth's geometric shape, orientation in space and gravity field.
- It supports accurate mapping, navigation, surveying and Earth observation.



## Objectives of Geodesy

- Determine the Earth's shape and size
- Establish coordinate systems
- Measure gravity field
- Monitor Earth's crustal movements and sea-level changes

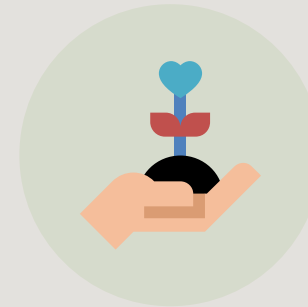


# Branches of Geodesy



## Geometric Geodesy

- Focuses on determining the size, shape, and position of the Earth using geometric methods.
- Ignores gravity variations; deals only with geometric relationships.
- Involves establishing and maintaining geodetic datums (e.g., WGS84, NAD83).
- Uses coordinate systems: latitude, longitude, ellipsoidal height and Cartesian (X, Y, Z).
- Techniques include:
  - Triangulation
  - Trilateration
  - Leveling
  - GNSS (Global Navigation Satellite Systems)
- Essential for mapping, navigation, and surveying.



## Physical Geodesy

- Studies the Earth's gravity field.
- Defines the geoid (mean sea level surface).
- Converts ellipsoidal heights to Orthometric heights.
- Uses tools like gravimeters and satellite missions (e.g., GRACE, GOCE).
- Important for height systems, sea level and mass movement studies.



## Satellite Geodesy

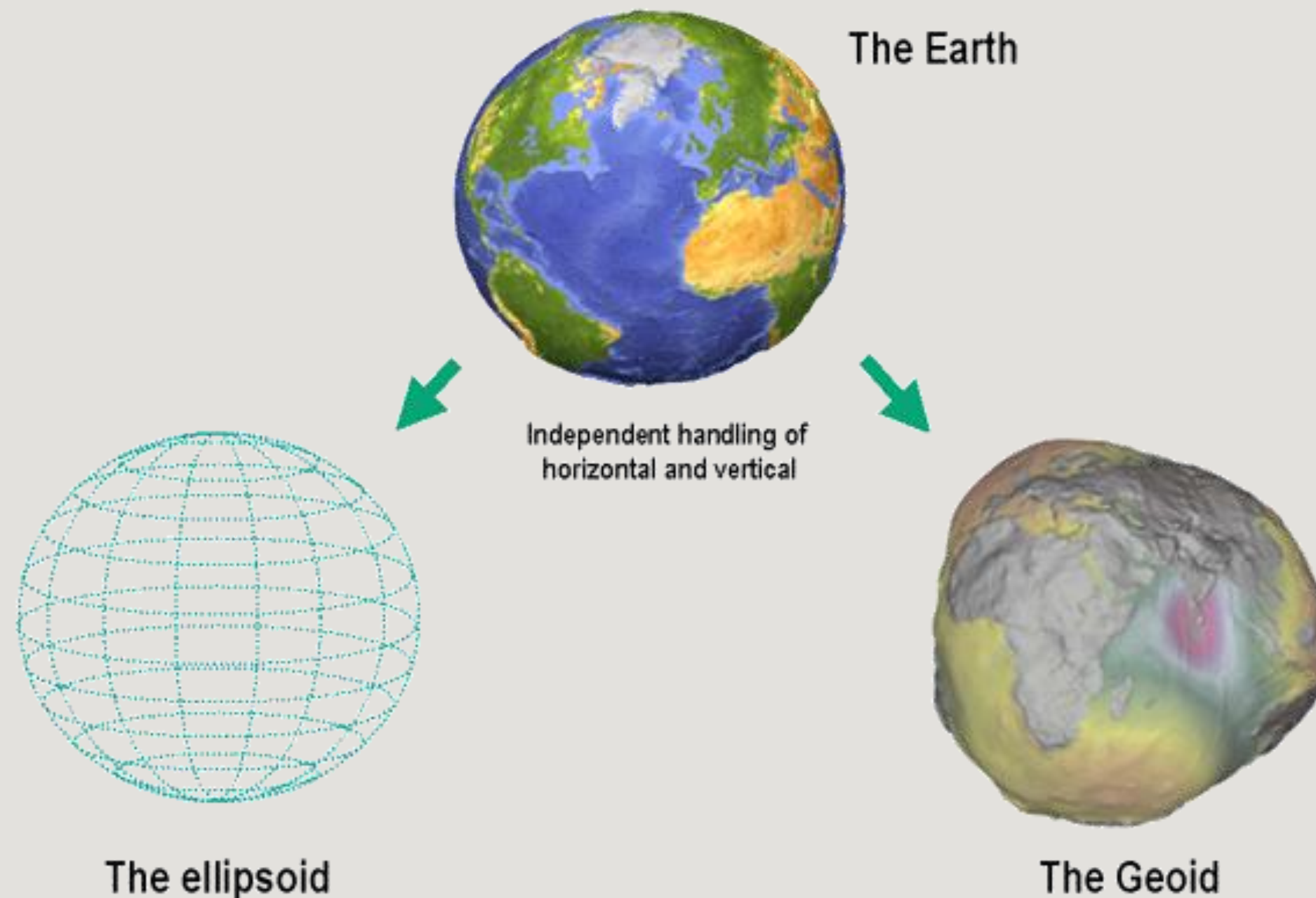
- Uses artificial satellites to measure the Earth's shape, position and gravity field.
- Enables global coverage and high accuracy.
- Main techniques:
  - GNSS (e.g., GPS, GLONASS, Galileo)
  - Satellite Laser Ranging (SLR)
  - CHAMP, GOCE
  - VLBI



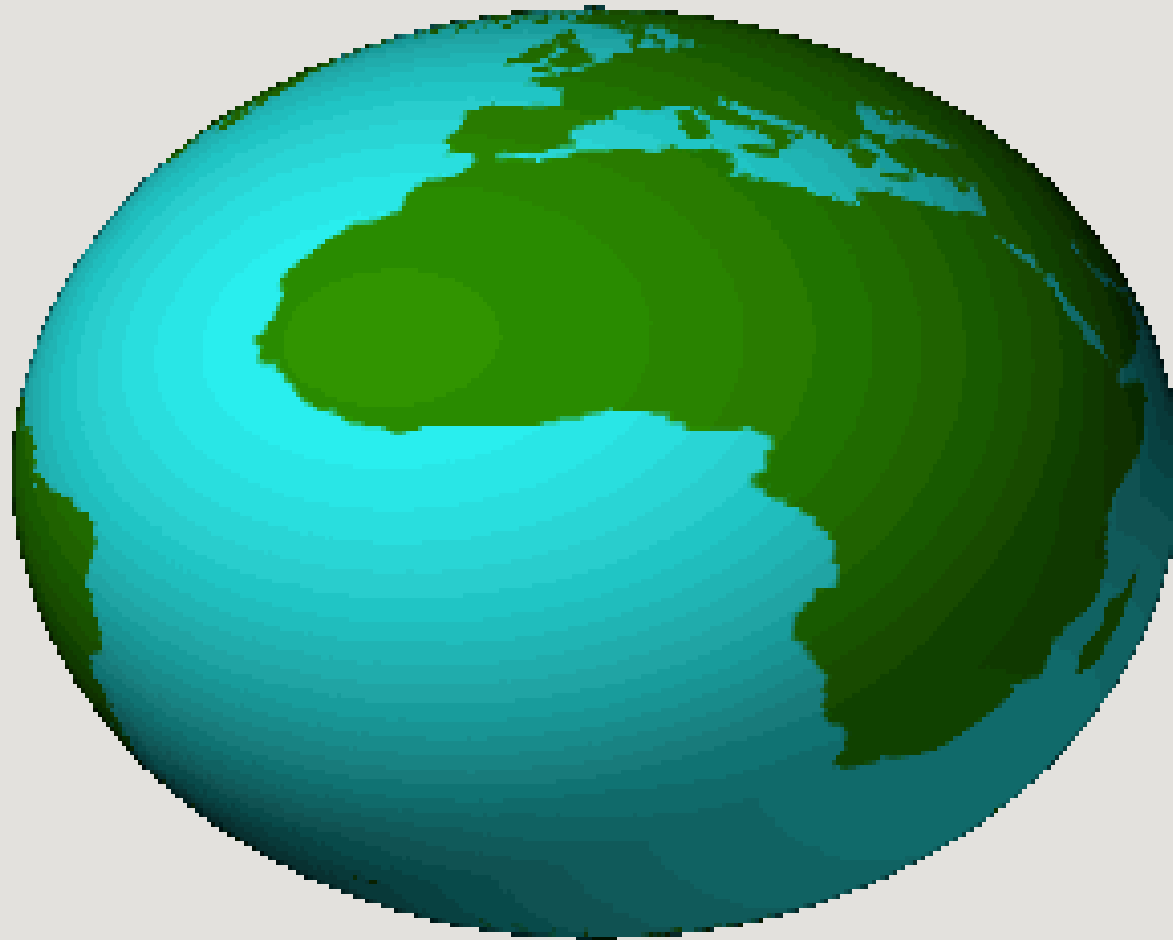
# Shape of the Earth

## Irregular Surface of the Earth

- The Earth's surface is uneven due to mountains, valleys and varying elevations (**Geoid**).
- It can be best **approximated** by a smooth, mathematical shape called an **ellipsoid**.



# Earth's Surface and Ellipsoid approximations



## What is an Ellipsoid?

A **geometric model** that closely resembles the shape of the Earth, though slightly flattened at the poles.

Defined by two key parameters:

### 1. Semi-major Axis ( $a$ )

- The **longest radius** of the ellipsoid, representing the equatorial radius.

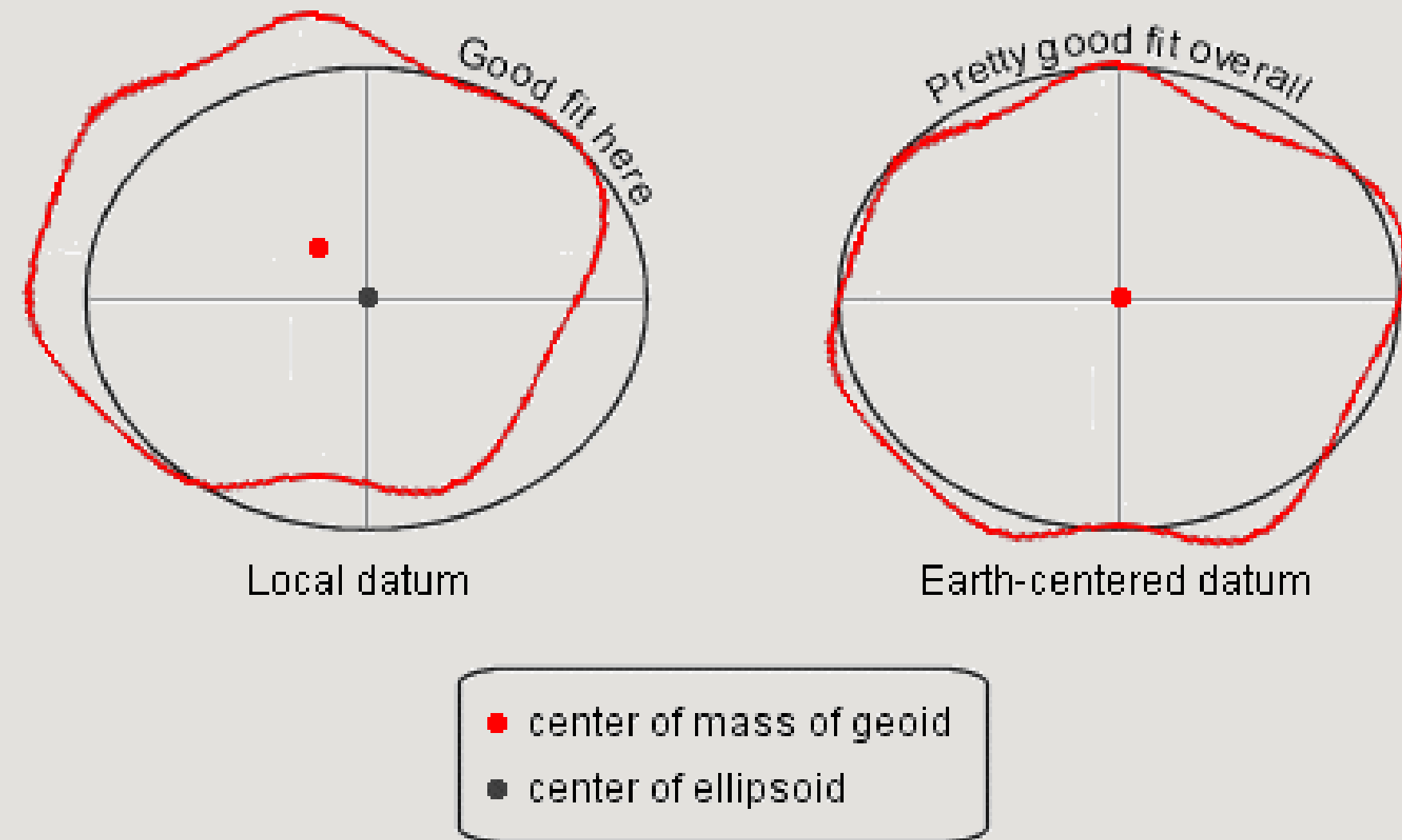
### 2. Eccentricity ( $e$ ) / Flattening ( $f$ )

- **Eccentricity** measures how much the shape deviates from being a perfect sphere.
- **Flattening** ( $f$ ) is the ratio of the difference between the equatorial and polar radii to the equatorial radius:

- where  $a$  is the semi-major axis and  $b$  is the semi-minor axis (polar radius).

$$f = \frac{a-b}{a}$$

# Local & Global Horizontal Datums



## 1. Local Horizontal Datum

**Definition:** A horizontal datum that is tied to a specific region or country, providing accurate geodetic coordinates for local use.

**Example:** Indian Datum (Kalianpur 1975), NAD83 (North American Datum 1983).

### Features:

- Best suited for small areas.
- Less accurate when applied globally due to local variations.
- Commonly used for national mapping, engineering, and construction.

## 2. Global Horizontal Datum

**Definition:** A geodetic system that provides a consistent framework for global use.

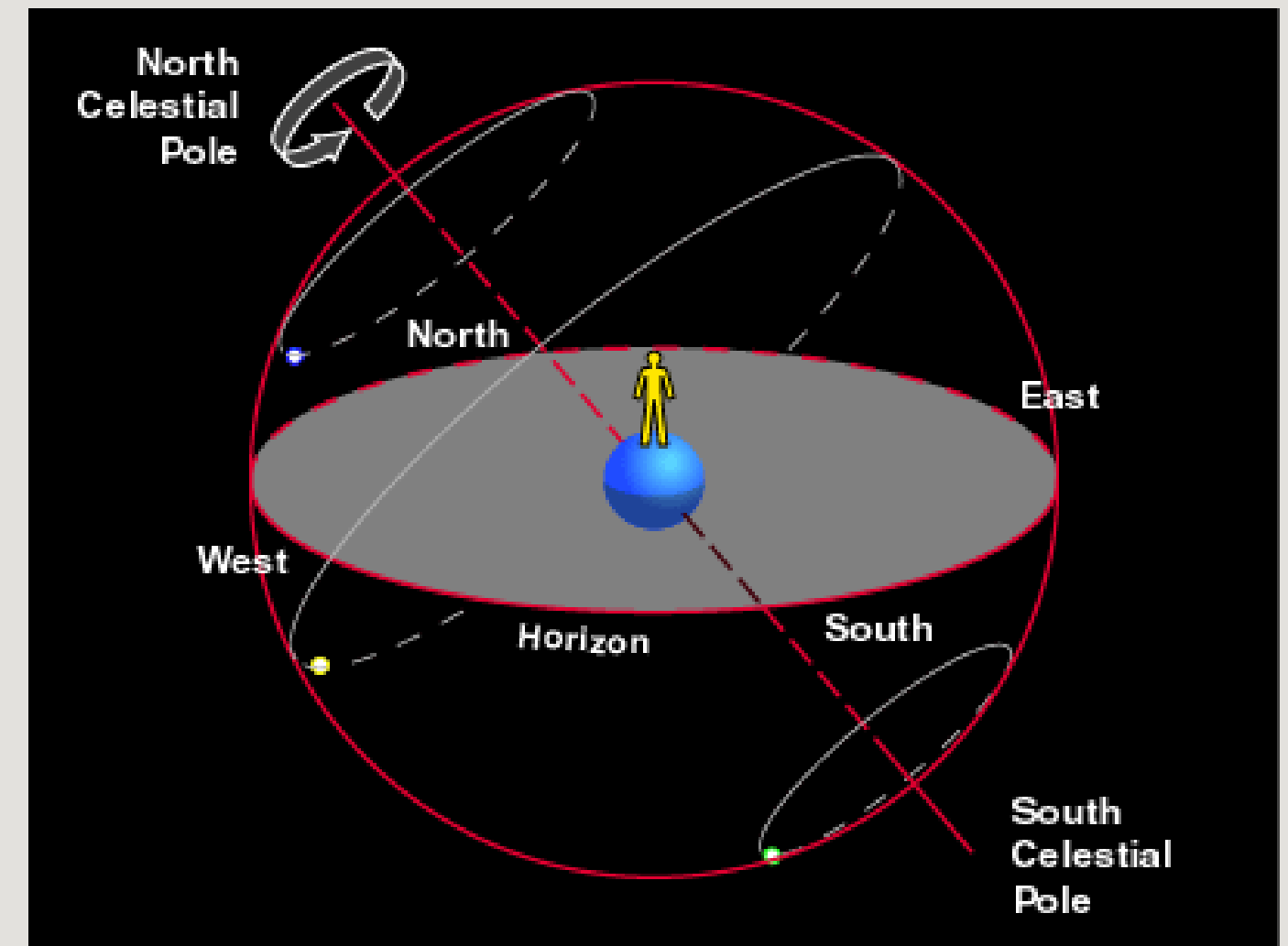
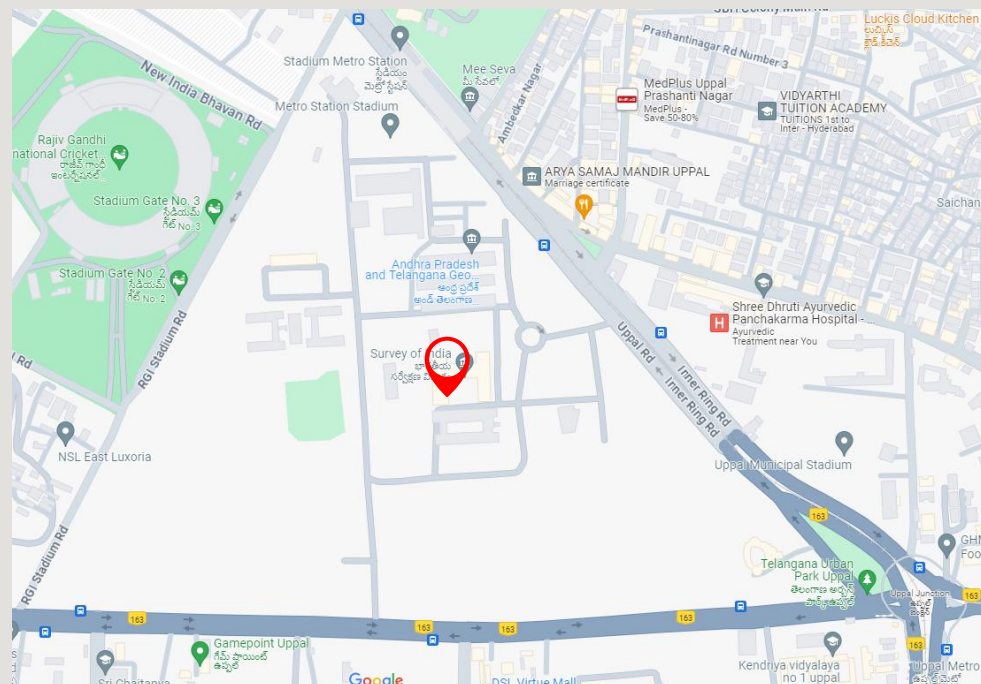
**Example:** WGS84 (World Geodetic System 1984), ITRF (International Terrestrial Reference Frame).

### Features:

- Used for global positioning systems (GPS) and satellite navigation.
- Provides uniform accuracy worldwide.
- Essential for global applications like aviation, maritime, and GPS navigation.

# Geographic Coordinate System

- **Latitude and Longitude:** Angular coordinates that are well-suited to the ellipsoidal shape of the Earth.
- Coordinates are expressed in **degrees, minutes, and seconds**, or other variations.
- **Positioning:** Coordinates are based on the angular distance from a known reference point.
- The reference point is where the **Greenwich Meridian** (Prime Meridian) and the **Equator** intersect.
- **Usage:** Latitude and Longitude is the primary coordinate system.



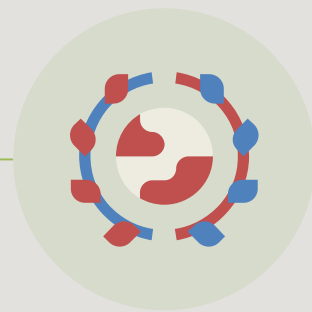
# Evolution in Survey of India



1767

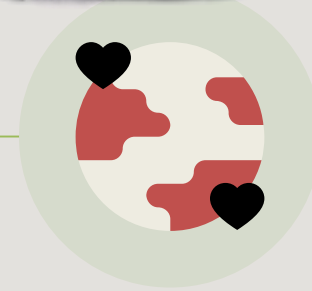
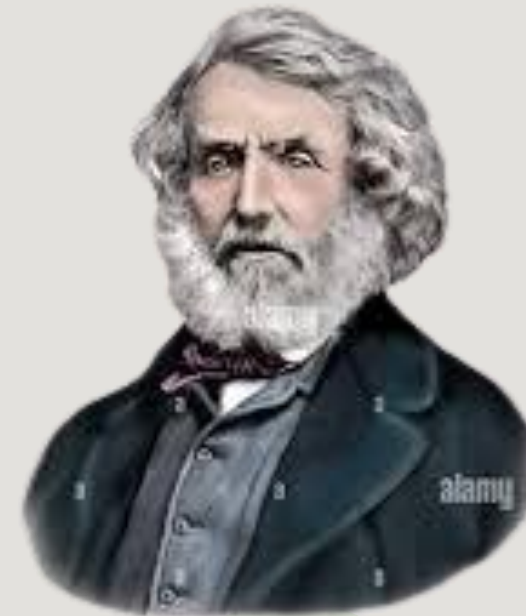
Set up by the  
**British East India**  
Company.

**Primary objectives:**  
Military mapping  
Land revenue  
assessment



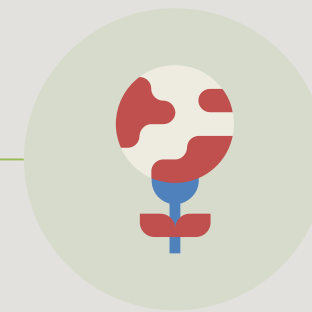
1802

**Great Trigonometrical  
Survey (GTS)**  
Precise measurement of arcs,  
base lines and Himalayan  
peaks



1830

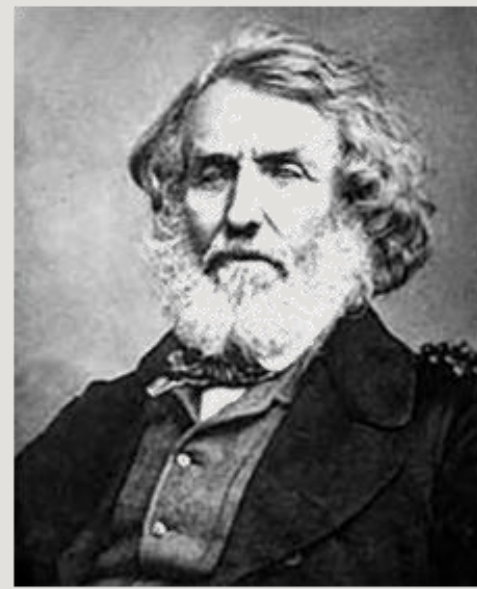
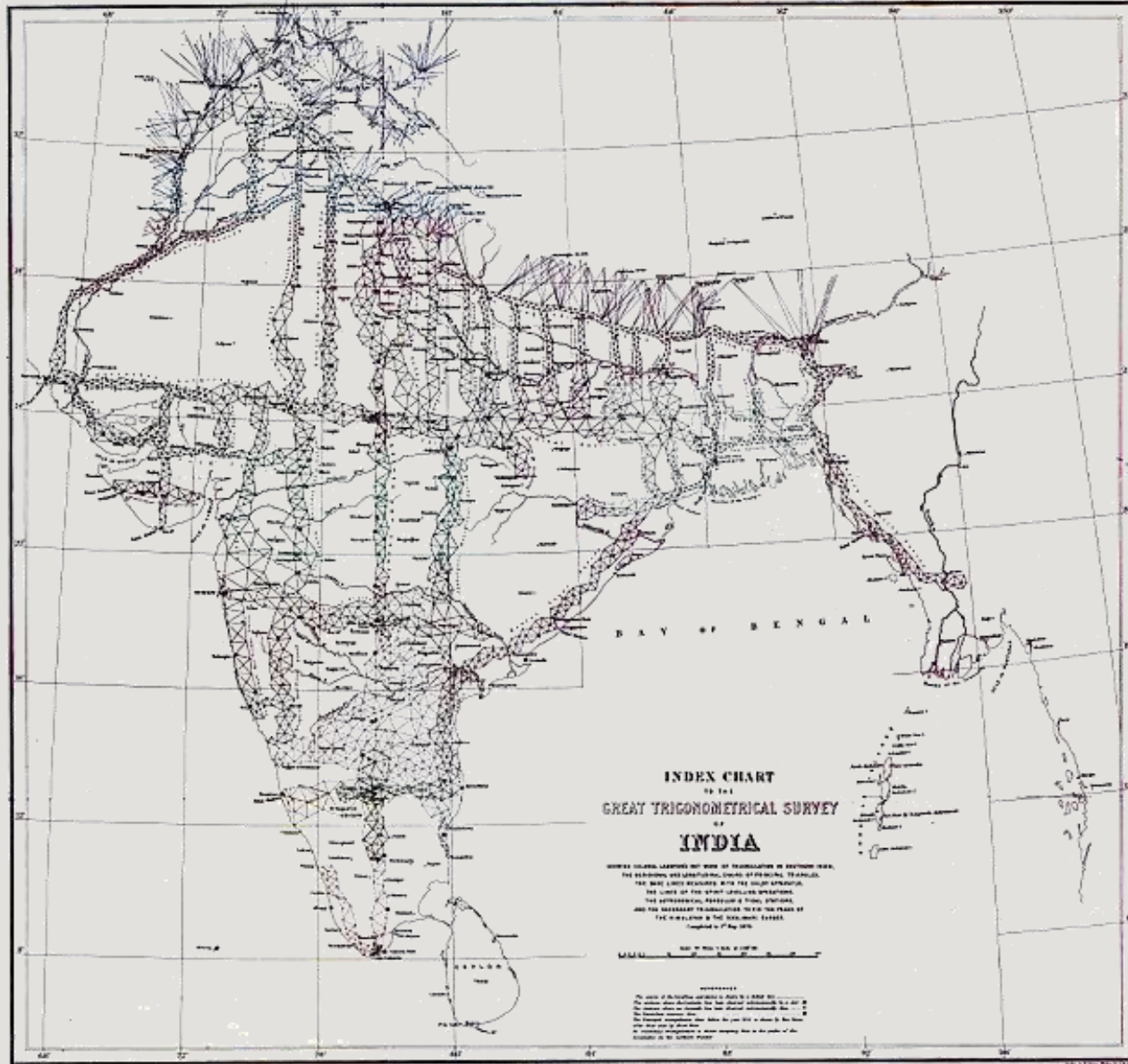
The **Everest Ellipsoid**  
was derived and  
published by  
**Sir George Everest.**



1909

Establishment of **Mean  
Sea Level (MSL)** as the  
Vertical Datum for India.

# Everest Ellipsoid and the Great Trigonometrical Survey (GTS)



## •GTS (Great Trigonometrical Survey):

- Initiated in **1802** under **Colonel William Lambton**.
- Involved highly accurate triangulation, covering over **2,400 km** from the southern tip of India to the Himalayas.

## •Everest Ellipsoid:

- Developed in **1830** by **Sir George Everest**.
- One of the **first ellipsoids** in the world designed specifically for India and surrounding regions.
- Designed to fit the curvature of the Earth in the Indian subcontinent, based on local measurements.

## •Ellipsoid Calculation:

- The **semi-major axis** of the Everest ellipsoid is **6,377,276.345 meters**.
- Flattening is **1/300.8017**—specific to the Indian region.

## •Impact and Usage:

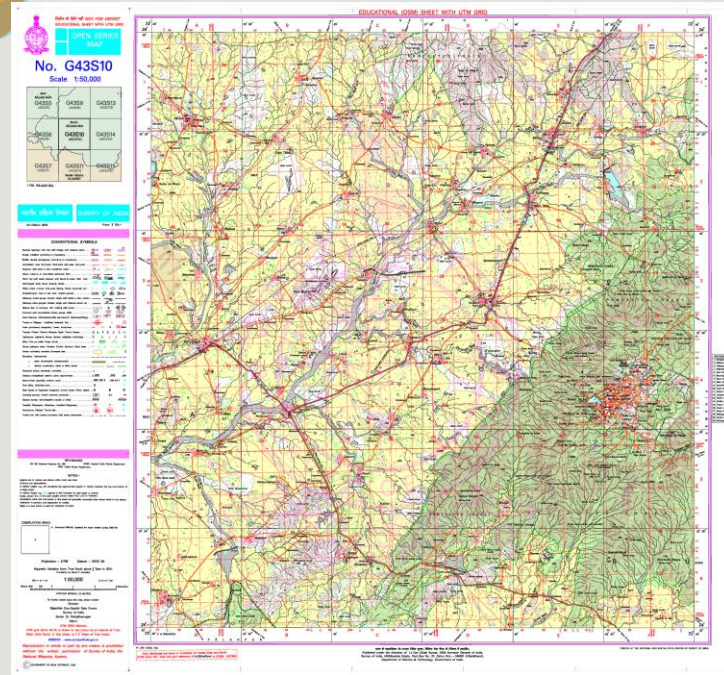
- Used for **over 150 years** as the primary geodetic reference in India.
- Gradually replaced by more globally fitting ellipsoids like **WGS84**, but Everest Ellipsoid still has historical importance in **local geodetic surveys**.



# Thoppikaran - the hat man of Thanjavur temple

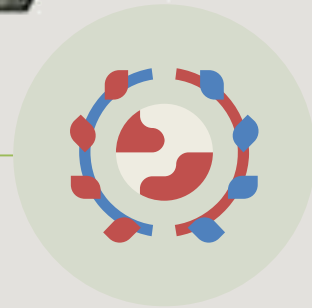
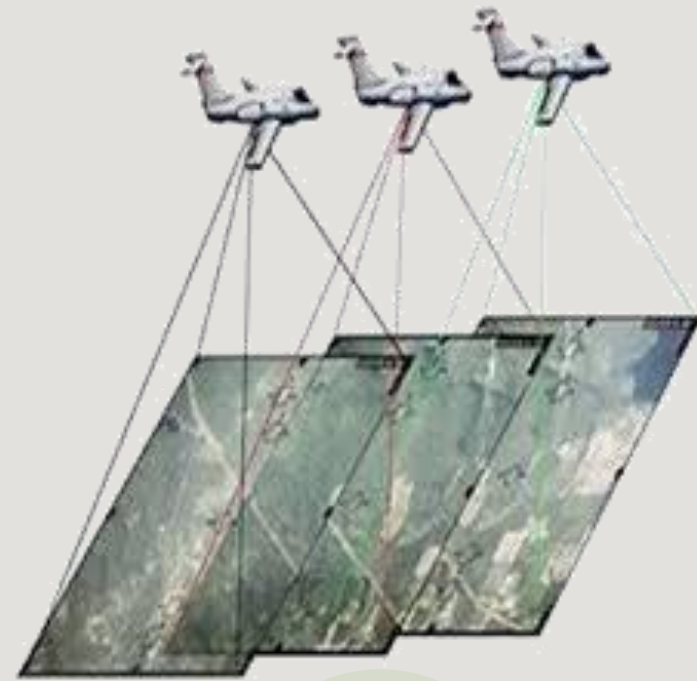


# Evolution in Survey of India



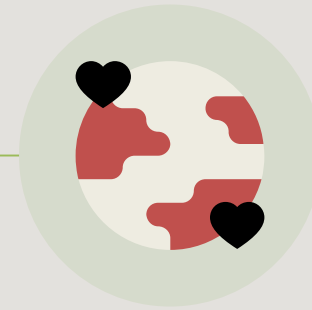
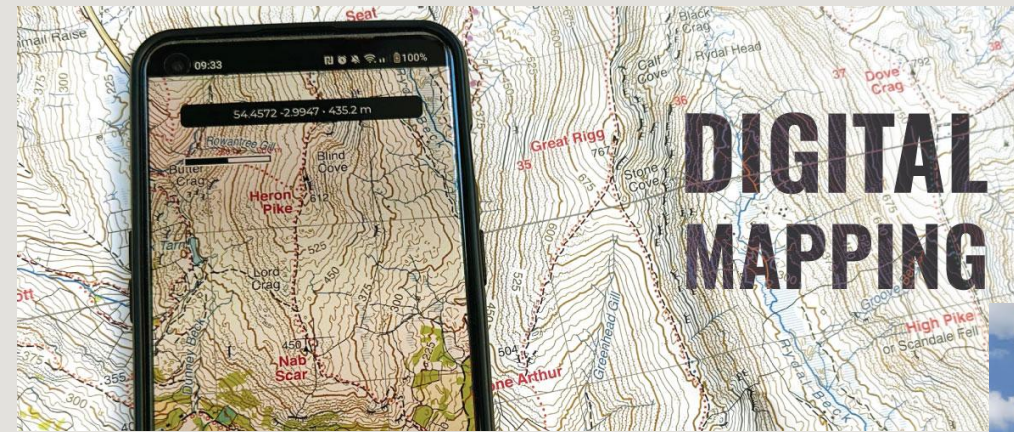
Early 1900s

Survey of India began standardizing **topographic map** production across India, especially using the **1 : 63,360 (1" = 1 mile)** scale.



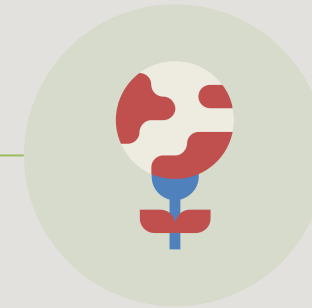
1960s

Began **aerial surveys** at national scale using Indian Air Force support.



1990s

Use of **GPS**, electronic distance measuring equipment (**EDM**) and **digital cartography** introduced.



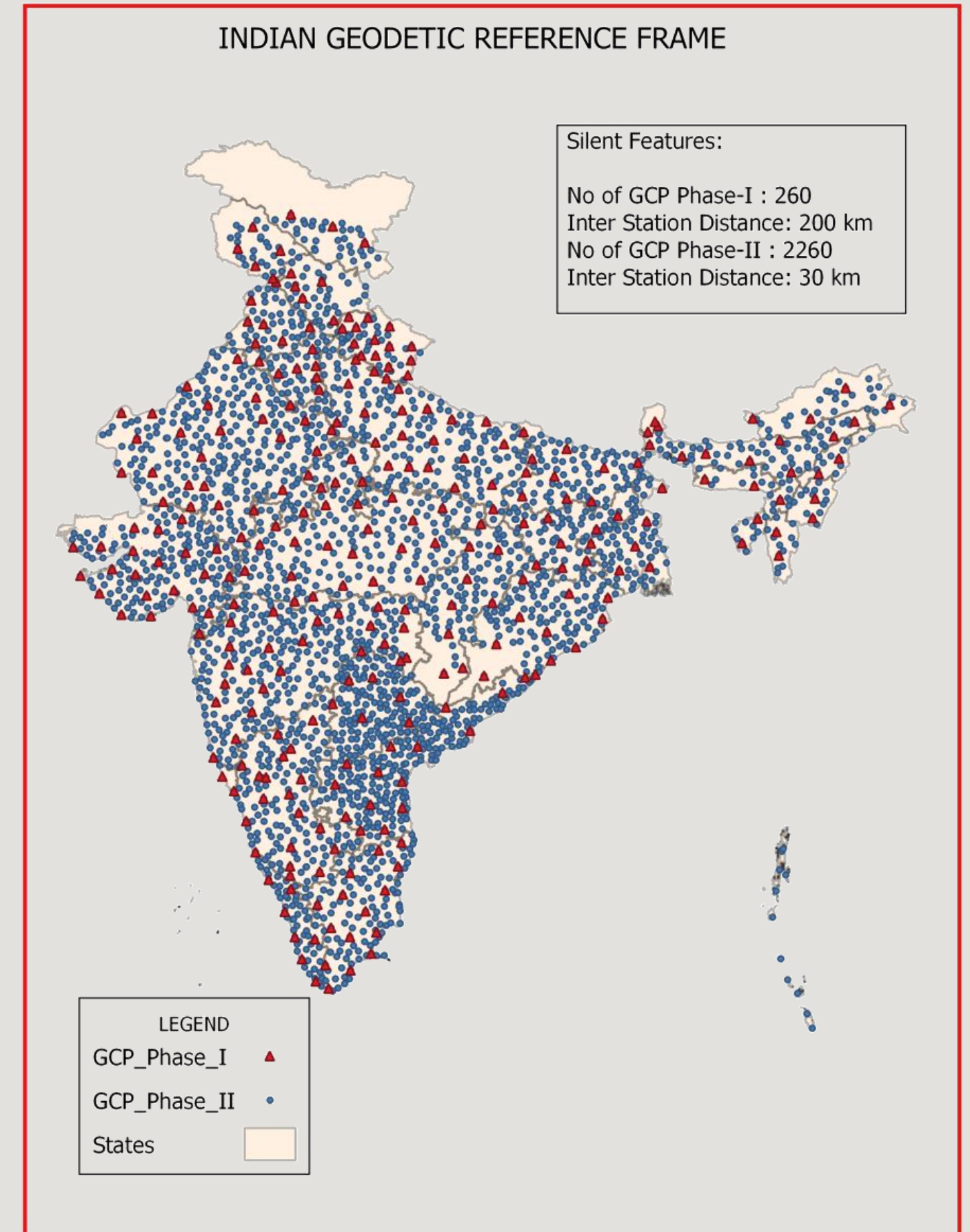
2021

Launch of **Continuously Operating Reference Station (CORS)** network

# National Spatial Reference Frame

- In National Map policy 2005, Survey of India has adopted **Geo-centric Coordinate System** WGS 84 Datum.
- GPS Survey oriented, Indian Geodetic Reference frame was realized by establishing Ground Control Points all over India
- **Ground Control Points (GCPs):**
  - **First Phase:** 260 GCPs spaced 250-300 km apart.
  - **Second Phase:** 2260 GCPs spaced 25-30 km apart.

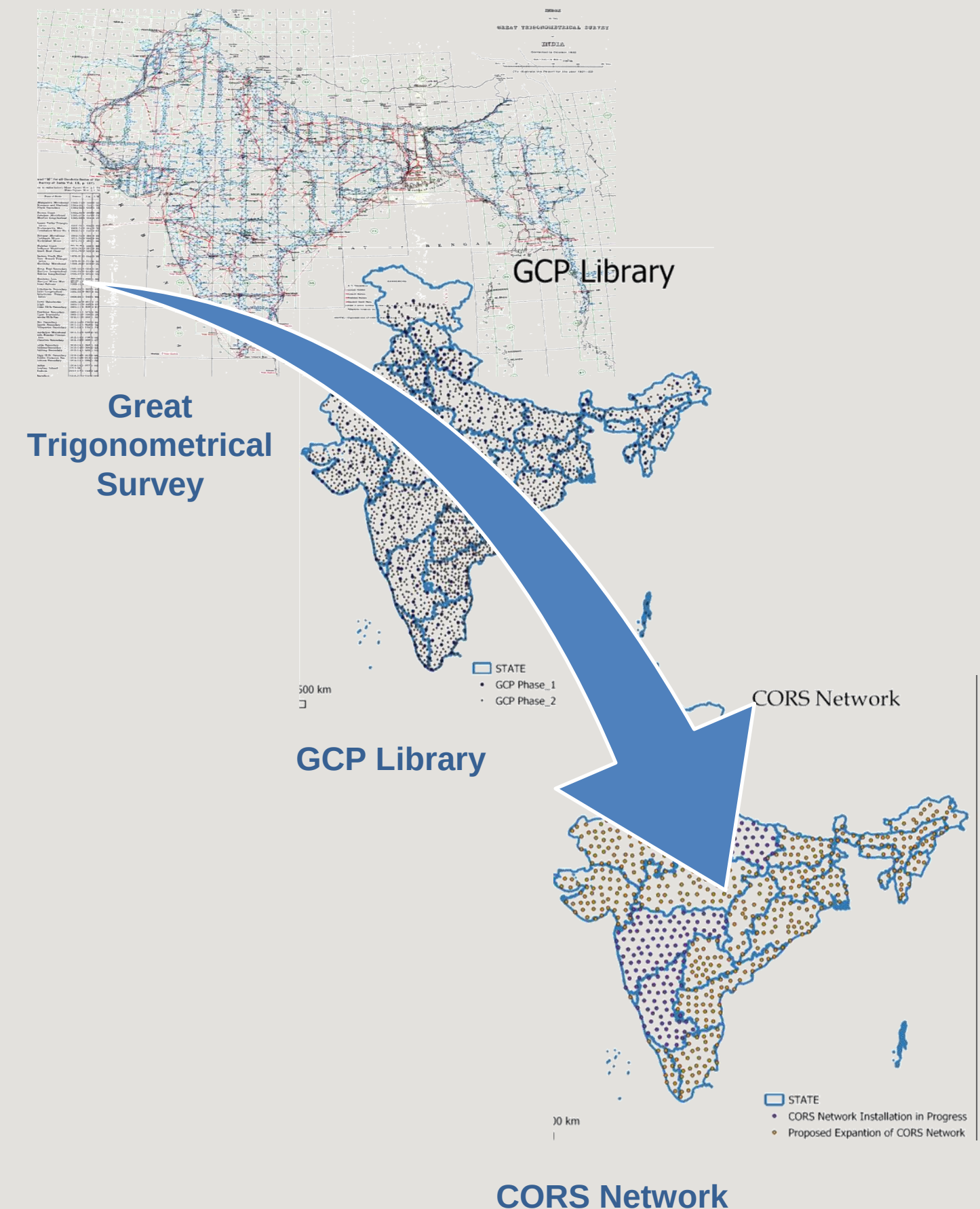
**Becoming obsolete due to aging and anthropological activities**



# National Spatial Reference Frame

- Requirements of NSRF

- **Well-defined**
  - in well conditioned geometries and distributed over all geotectonic regions
- **Highly accurate**
  - Precisely measured Baseline to enable mm to cm level accuracy
- **Easily accessible**
  - Common, hassle free platform
- **Long term stable**
  - By ground fixed Monumental stations
- **Coordinates, as well as velocities**
  - To accommodate long term geotectonic movements,
  - To be used for Earth system observations
- **Co-Located**
  - To be measured with other sensors, to measure precision and redundancy
  - To be linked with other Datums like Vertical, Gravity, Tidal etc



# Basics of GNSS



## Definition of GNSS

✓ GNSS stands for Global Navigation Satellite System, which provides positioning, navigation and timing (PNT) services globally. It consists of multiple satellite constellations that help users determine their precise location anywhere on Earth.

## Difference between GNSS and GPS

✓ GPS (Global Positioning System) is a specific GNSS developed by the United States. GNSS is a broader term that includes multiple satellite systems operated by different countries.



## GPS

- Since 1980s
- 31 Sats
- GPS III Launch Dec 2018



## GLONASS

- Refurb 2012
- Operating well with 24 sats



## Galileo

- Full capability in 2022
- 30 sats



## Beidou

- Full capability in 2020
- 33 sats for BD-3

## Regional Systems



Japan - QZSS



India - NAVIC



# World Geodetic System of 1984 (WGS84)

**WGS84** is the global reference system used by GPS to define positions on Earth.

It tells us the size and shape of the Earth and provides a standard coordinate system to measure locations accurately.

It was created in 1984 by the U.S. Defense Mapping Agency using the best technology and data available at that time.

It uses a smooth, mathematical shape of Earth called an ellipsoid:

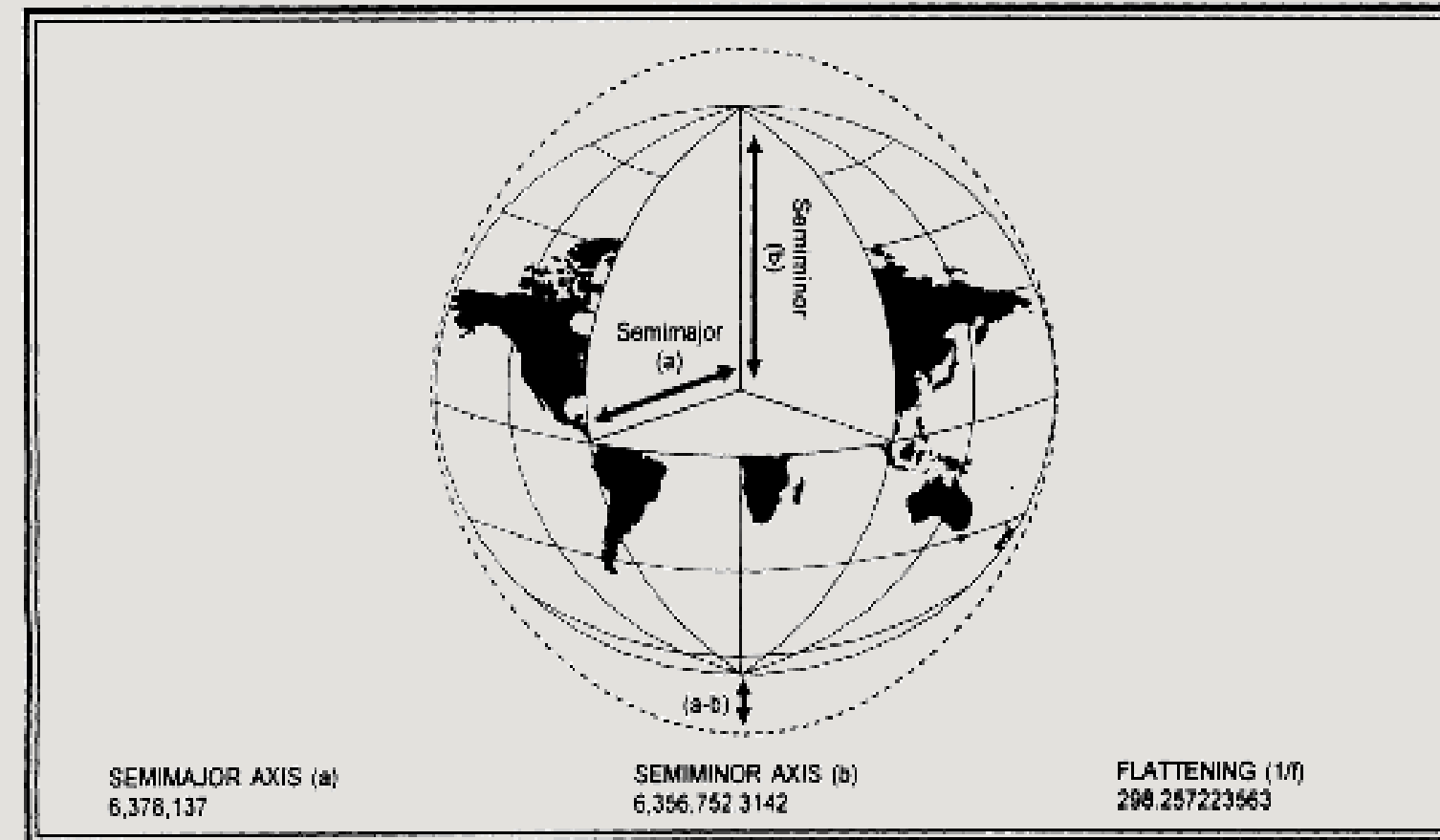
Equatorial radius (a) = 6,378,137 meters

Flattening (f) =  $1 / 298.2572236$

This model is centered at the Earth's center (geocentric) and was based on data from around 1,500 ground stations.

WGS84 helps ensure that everyone around the world uses the same reference for maps, navigation and GPS.

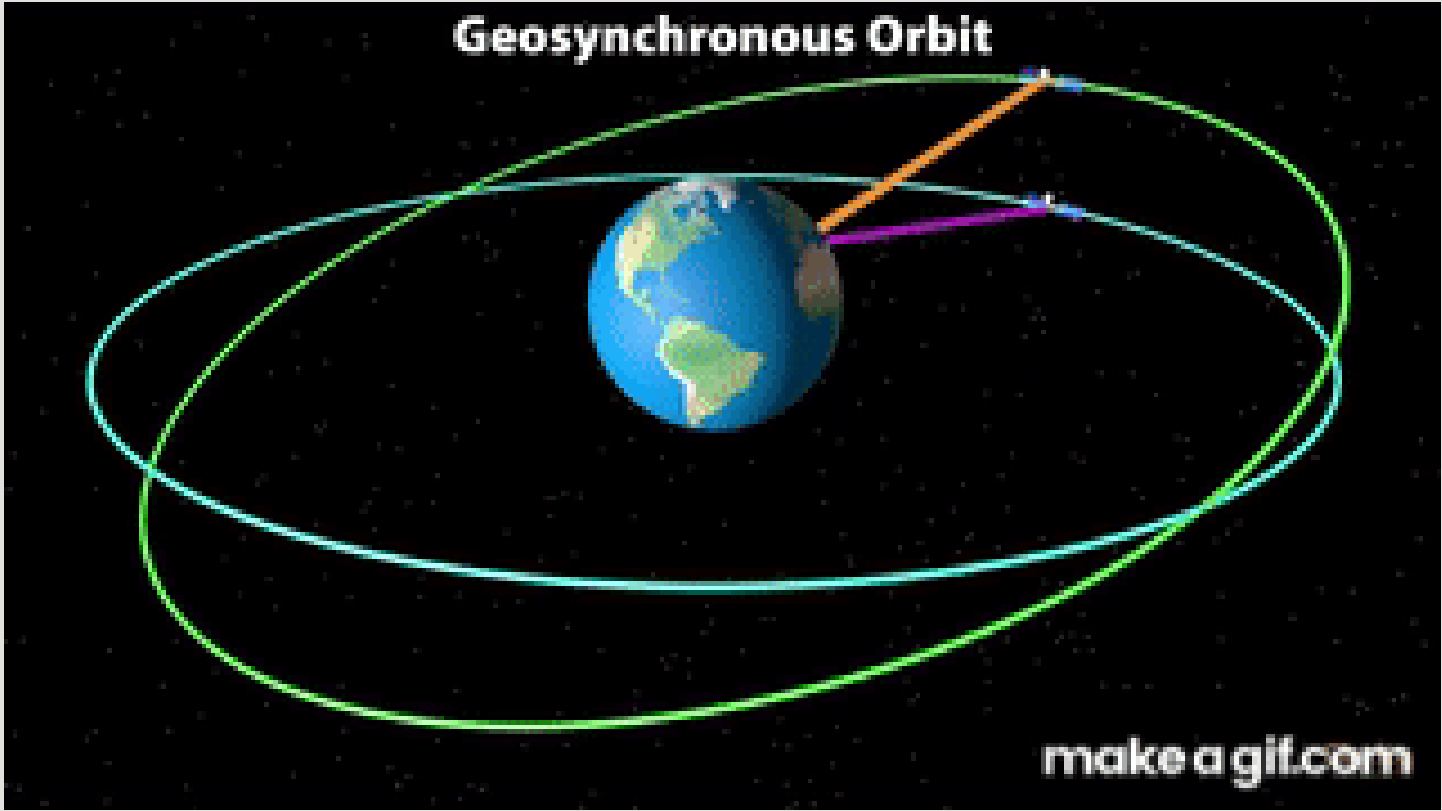
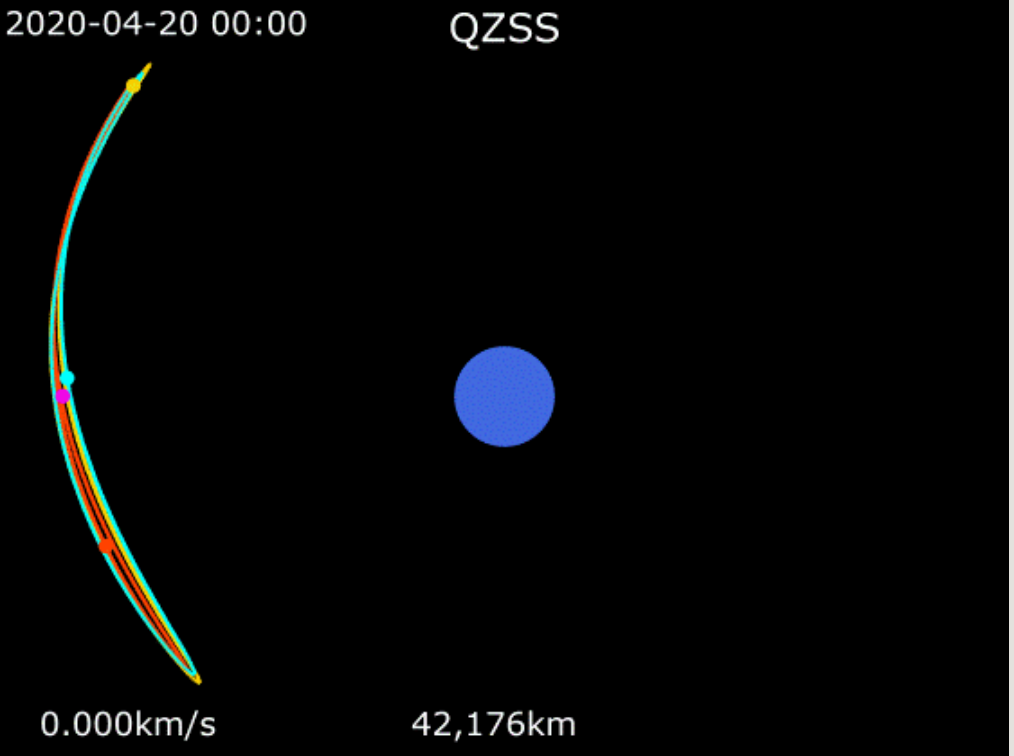
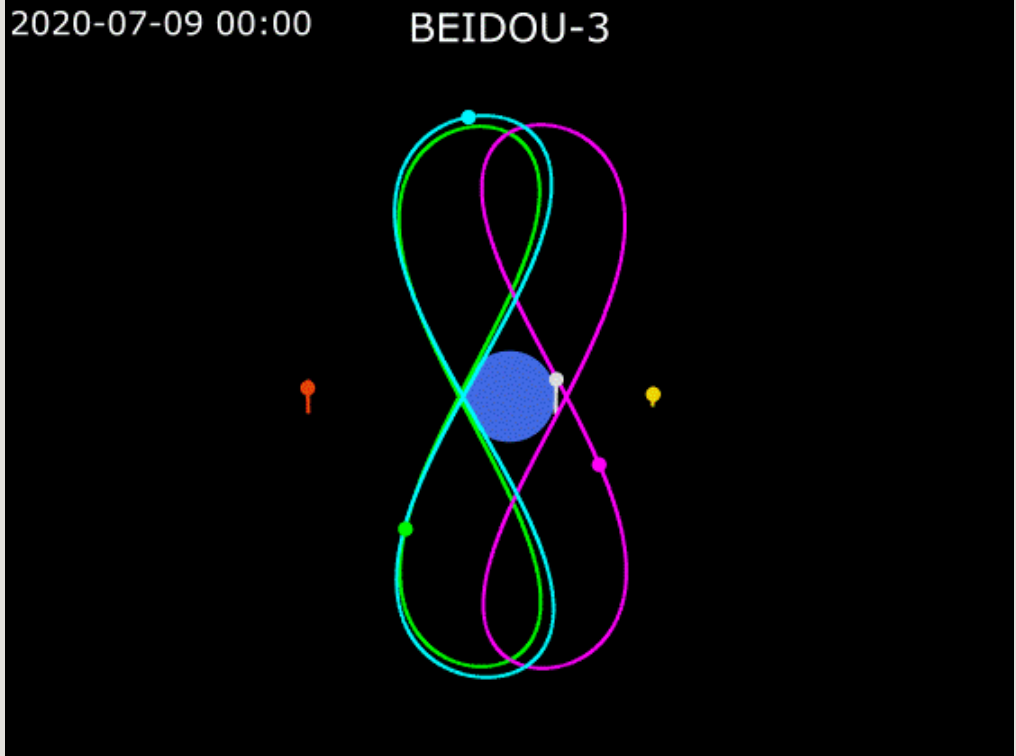
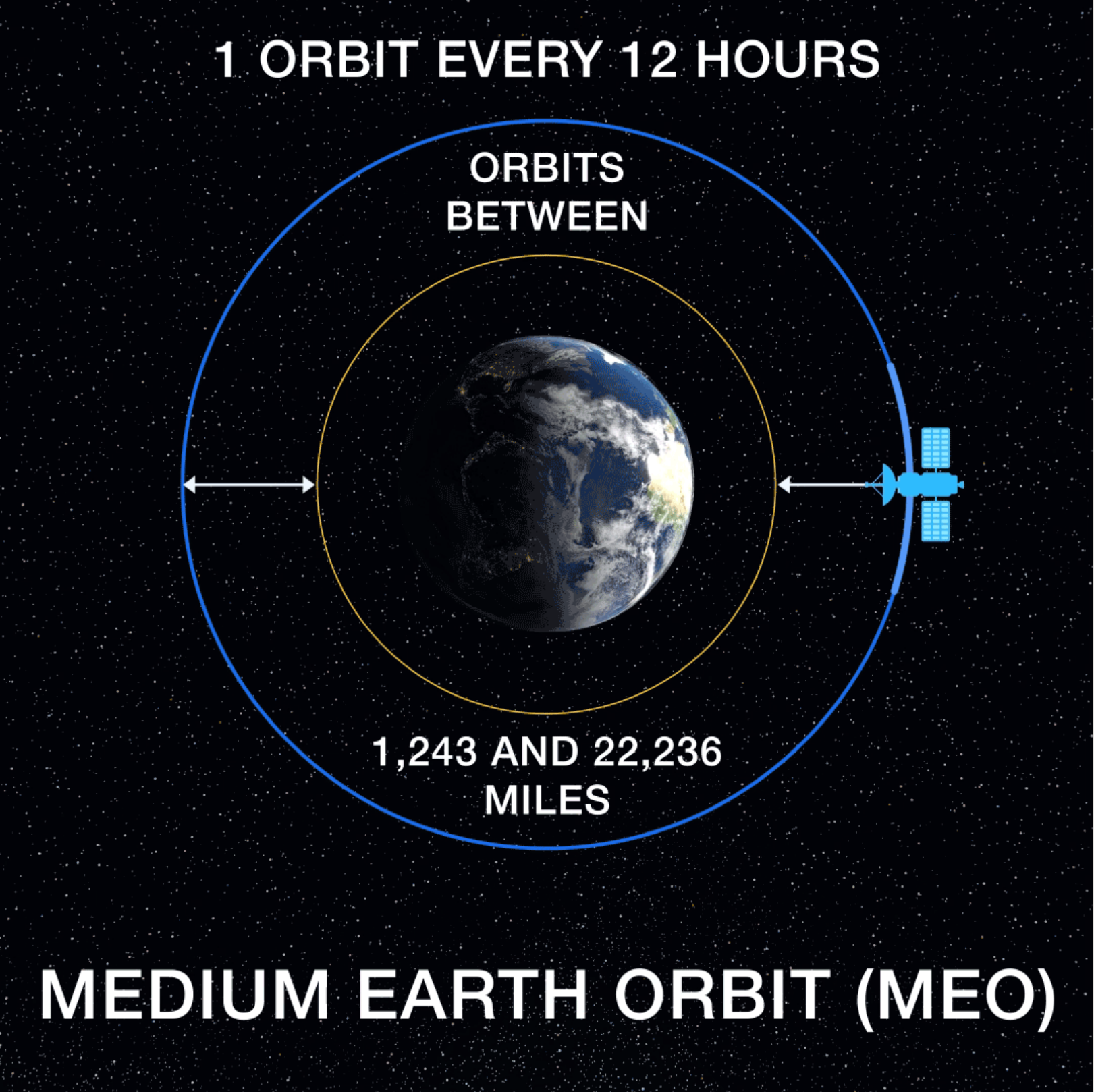
Figure E-1. World Geodetic System 1984



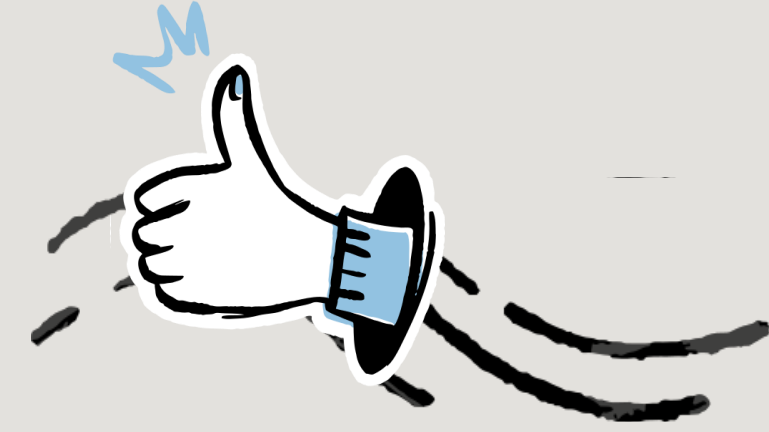
# What is GNSS

| GNSS System      | Coverage Type                     | Current Satellites                 | Notable Details                                          |
|------------------|-----------------------------------|------------------------------------|----------------------------------------------------------|
| GPS (USA)        | Global                            | ~31 in MEO                         | Oldest GNSS, established global coverage                 |
| GLONASS (Russia) | Global                            | 24–26 in MEO                       | Better high-latitude coverage due to orbital inclination |
| Galileo (EU)     | Global                            | ~26–30 in MEO                      | Civilian-focused, interoperable with GPS/GLONASS         |
| BeiDou (China)   | Global                            | ~35 across MEO, IGSO, GEO          | Multi-orbit design for global & regional coverage        |
| REGIONAL SYSTEMS |                                   |                                    |                                                          |
| NavIC (India)    | Regional<br>(India + surrounding) | ~8 in GEO/IGSO                     | Focused regional coverage, expanded with latest launches |
| QZSS (Japan)     | Regional<br>(Asia-Oceania)        | (3 in HIEO + 1 in GEO) 4 currently | Works in concert with GPS for improved regional accuracy |

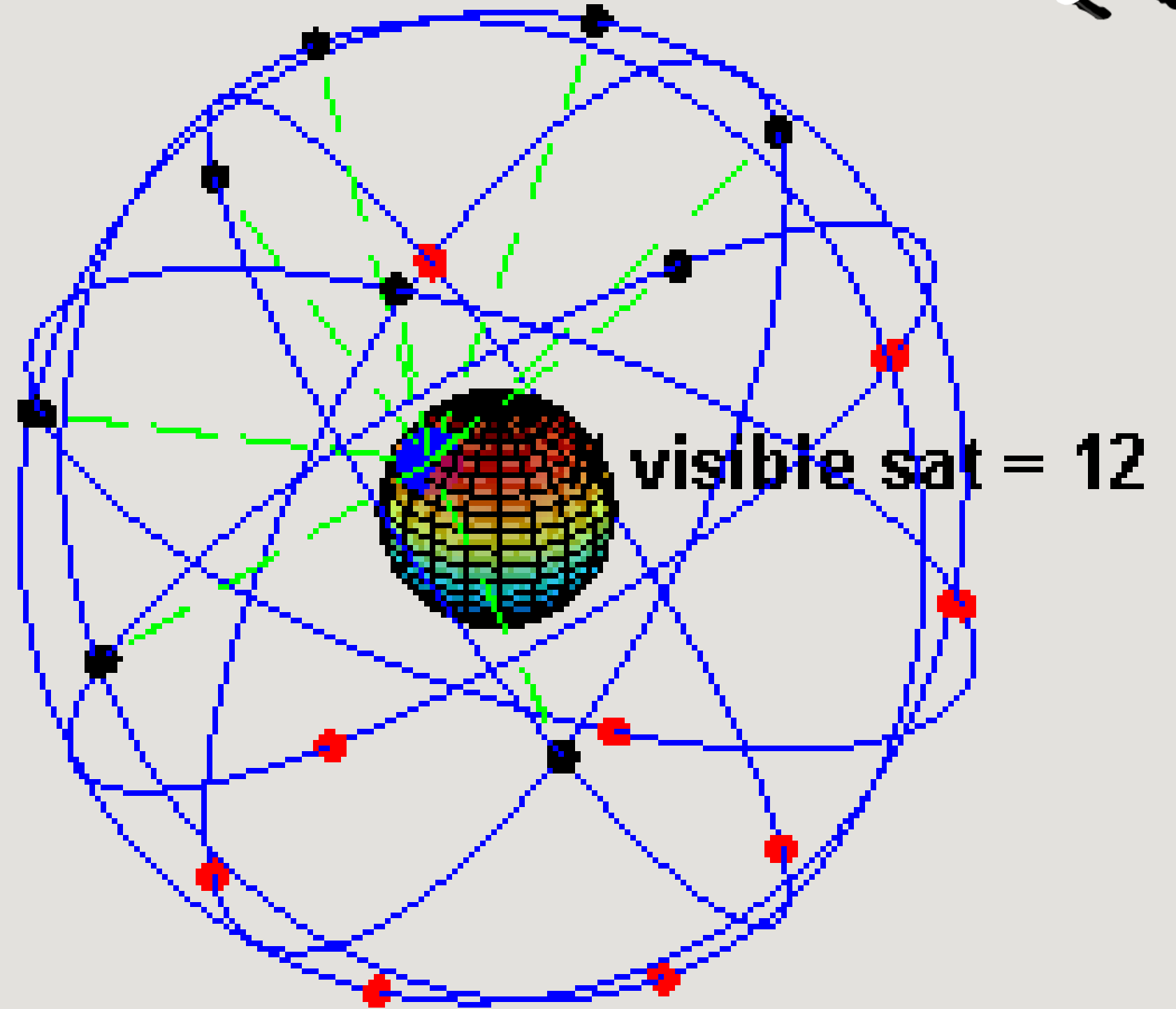
# MEO, GEO, HIEO & IGSO



# 01 Key Segments of GNSS



- **Space Segment:** Satellites in orbit transmitting signals.
- **Control Segment:** Ground stations monitoring satellite health and updating data.
- **User Segment:** GNSS receivers in devices like smartphones, survey instruments, and cars.



# Key Segments of GNSS

**Space Segment:** Consists of a constellation of satellites orbiting the Earth in precise paths. Each satellite continuously transmits signals containing:

- **Satellite ID**
- **Orbital position (ephemeris)**
- **Precise time from its atomic clock**

**Control Segment:** Network of ground-based control stations and antennas around the globe.

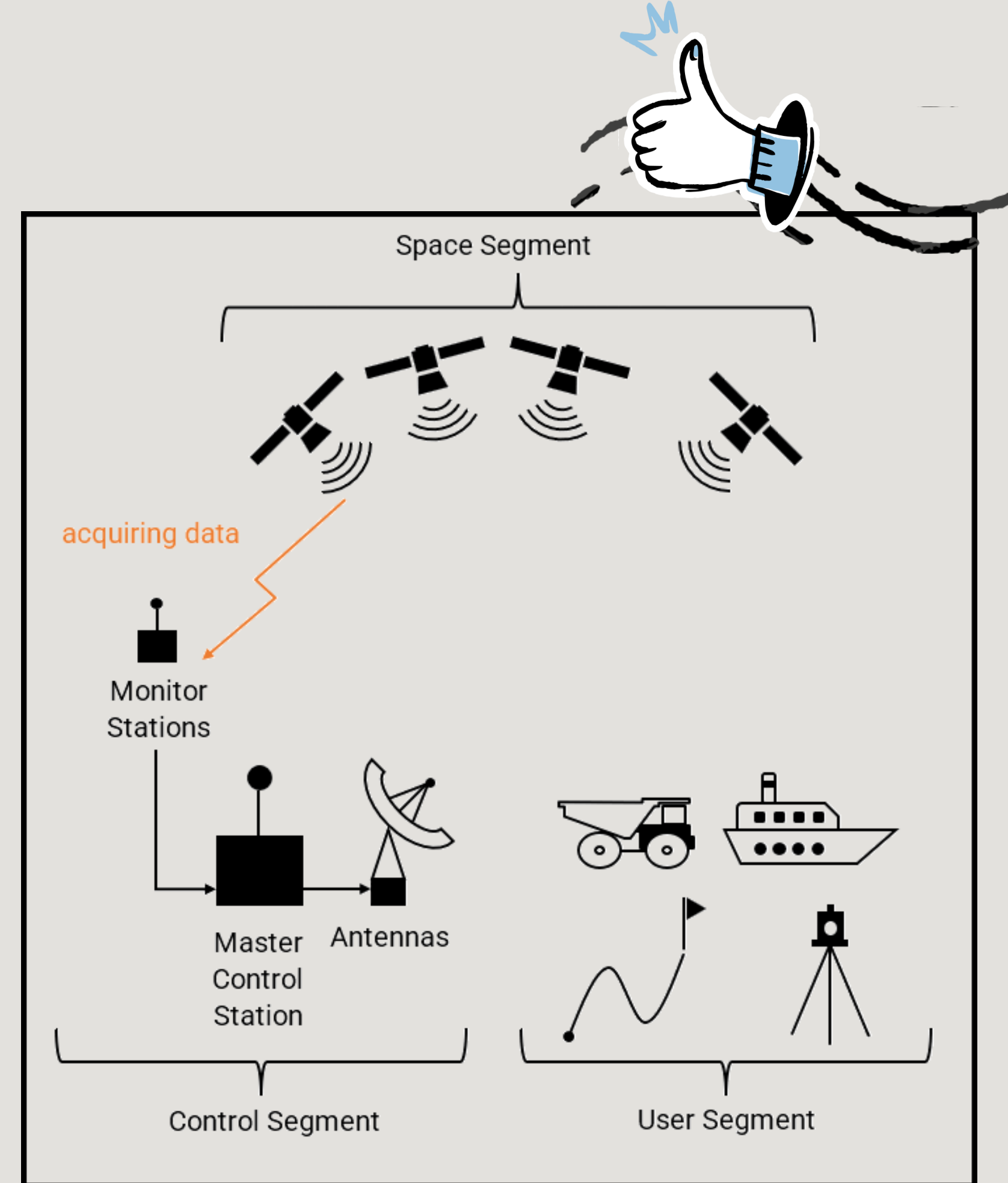
- **Monitor satellite health and status**
- **Track precise satellite orbits**
- **Update clock corrections and ephemeris data**
- **Upload navigation messages to satellites**

**User Segment:** includes any GNSS receiver—ranging from smartphones to high-precision survey instruments.

**For land surveying:** Survey-grade GNSS receivers track **multiple constellations (multi-GNSS)** and multiple frequencies (e.g., **L1, L2, L5**).

Capable of real-time corrections (RTK) or post-processing (Static/Rapid Static/PPK).

**GNSS receivers determine:** Position (latitude, longitude, height) Velocity and Time.



**Master Control Station (MCS)**

**One** Master Control Station—located near Colorado Springs, Colorado—responsible for managing satellite orbits, clocks, health, and the generation of navigation messages. This station is backed by an alternate (backup) Master Control Station to ensure continuity and reliability.

**Monitor Stations (MS)**

There are **16 Monitoring Stations** globally distributed. These include:

- **6 U.S. Space Force** (Air Force) dedicated stations
- **10 shared stations** operated by the National Geospatial-Intelligence Agency (NGA)

Together, they continuously track GPS satellites, providing essential data for orbit and clock corrections.

**Ground Antennas (GA)**

The system comprises 4 dedicated GPS ground antennas, co-located at these sites:

- Ascension Island
- Cape Canaveral
- Diego Garcia
- Kwajalein Atoll

Additionally, the system can leverage **7 Air Force Satellite Control Network (AFSCN)** remote tracking stations for enhanced communication and control support.



**Legend:**

● NGA Monitoring Stations

◆ Master Control Station

*(Schriever A.F.B. Colorado Springs, CO)*

◆ Alternate Master Control Station

◆ AFSCN Remote Tracking Station

◆ Air Force Monitoring Stations

★ Ground Antenna



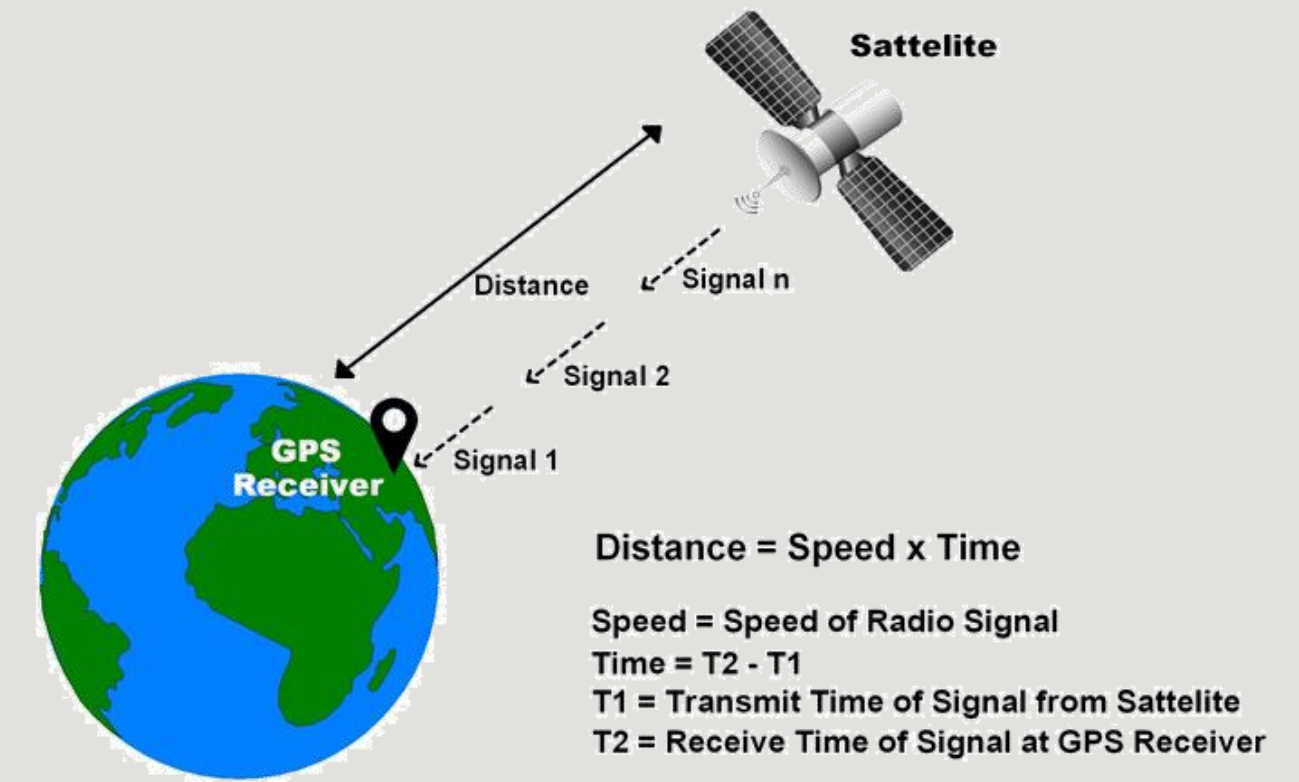
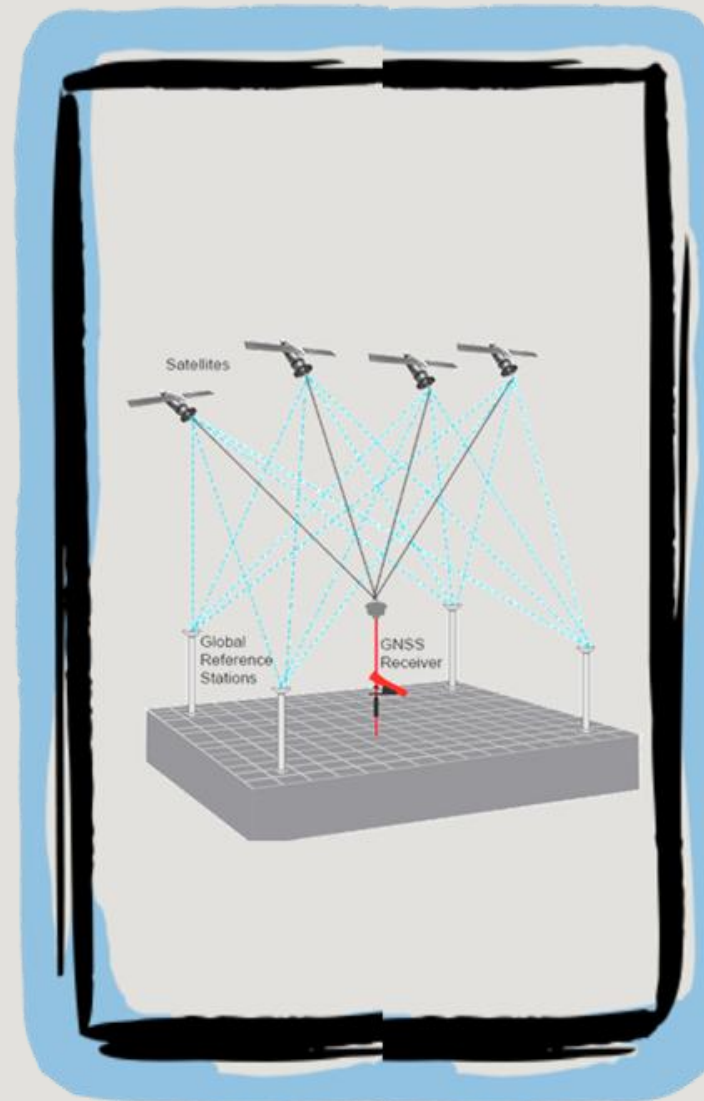
# How GNSS Works?

## 02

## Basic Working Principle of GNSS

- GNSS works by using satellite signals to determine a receiver's position on Earth. The process involves:
  - Satellites transmitting signals with precise time and orbital information.
  - Receivers calculating the time of signals to determine distance from each satellite.
  - Using at least four satellites to compute accurate latitude, longitude, altitude, and time (4D positioning).

$$\rho = c \cdot \Delta t$$



Equation 1:

$$\rho_1 = \sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2} + c \cdot b$$

Equation 2:

$$\rho_2 = \sqrt{(x - x_2)^2 + (y - y_2)^2 + (z - z_2)^2} + c \cdot b$$

Equation 3:

$$\rho_3 = \sqrt{(x - x_3)^2 + (y - y_3)^2 + (z - z_3)^2} + c \cdot b$$

Equation 4:

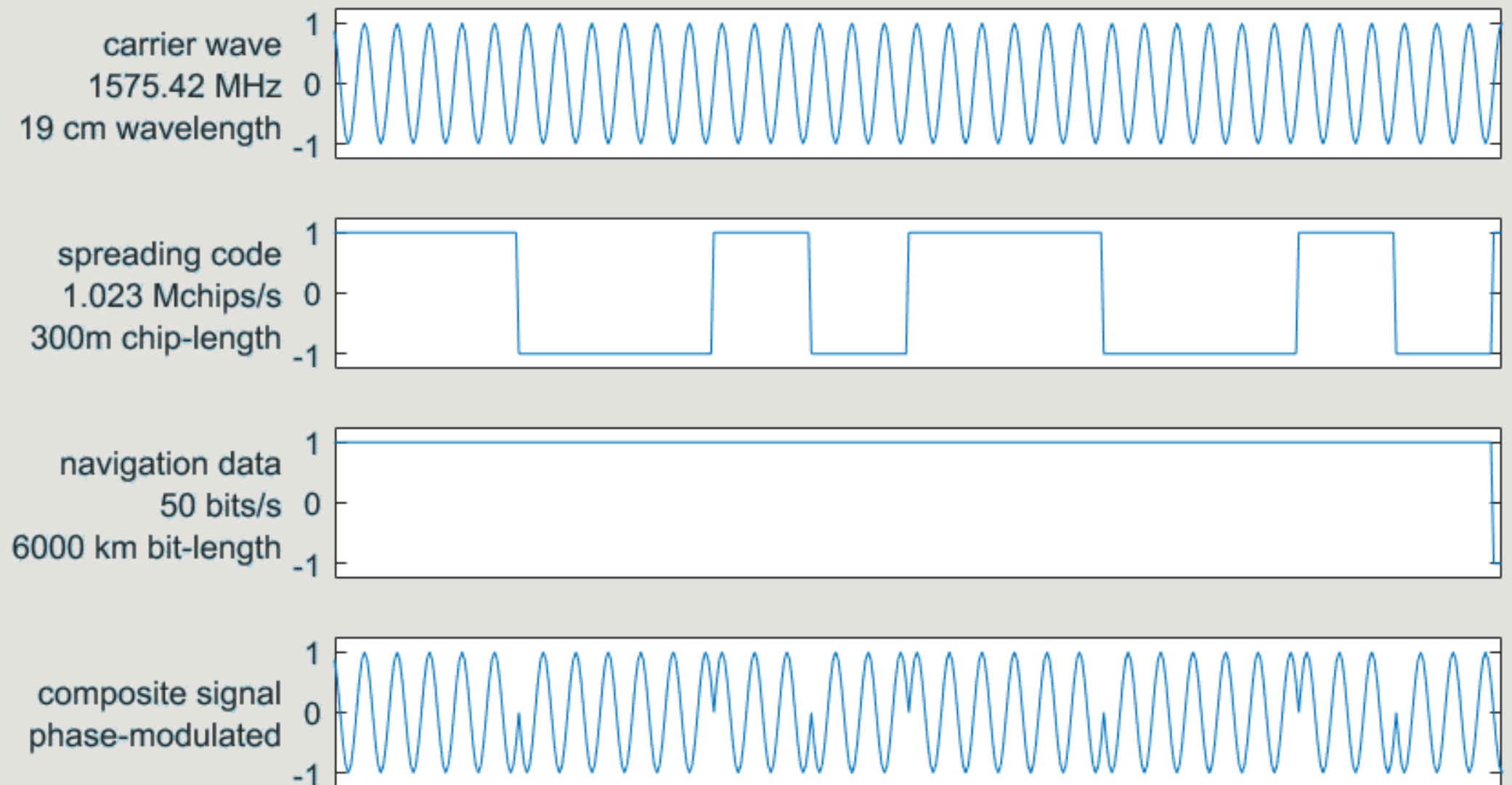
$$\rho_4 = \sqrt{(x - x_4)^2 + (y - y_4)^2 + (z - z_4)^2} + c \cdot b$$

# 03

## GNSS Signal Components

### 1. Carrier Signal

- A high-frequency sine wave transmitted by satellites.
- Used for precise distance (phase-based) measurements in RTK/NRTK.
- **Examples:**
  - GPS L1 (1575.42 MHz), L2, L5
  - Galileo E1, E5a, E5b



### 2. PRN (Pseudo-Random Noise Code)

- A unique digital code sequence assigned to each satellite.
- Allows receivers to identify which satellite is sending the signal.
- Enables precise pseudorange measurement by correlating sent and received codes.

### 3. Navigation Message

#### Carries vital satellite information:

- Orbit (ephemeris) & clock corrections
- Satellite health/status
- Time information:
- Essential for computing satellite positions and correcting receiver time bias.

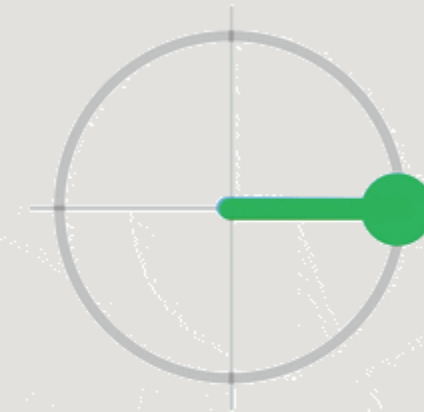
# GNSS Signal Components

04

## Carrier Waves

(Used for precise distance measurement)

- GNSS signals are transmitted on specific frequencies.
- Used in carrier-phase measurements for high-precision positioning.
- Helps achieve millimeter-level accuracy in RTK and PPP techniques.



Difference:  $0^\circ$



In Phase

## Key Carrier Frequencies in GNSS:

- GPS:** **L1** (1575.42 MHz), **L2** (1227.60 MHz), **L5** (1176.45 MHz)
- Galileo:** **E1** (1575.42 MHz), **E5a** (1176.45 MHz), **E5b** (1207.14 MHz)
- GLONASS:** **L1** (1602 MHz), **L2** (1246 MHz)
- NAVIC (Navigation with Indian Constellation)** are: **L5 Band:** 1176.45 MHz (Primary civilian frequency)  
**S Band:** 2492.028 MHz (Unique to NAVIC, civilian and military)

# GNSS Signal Components

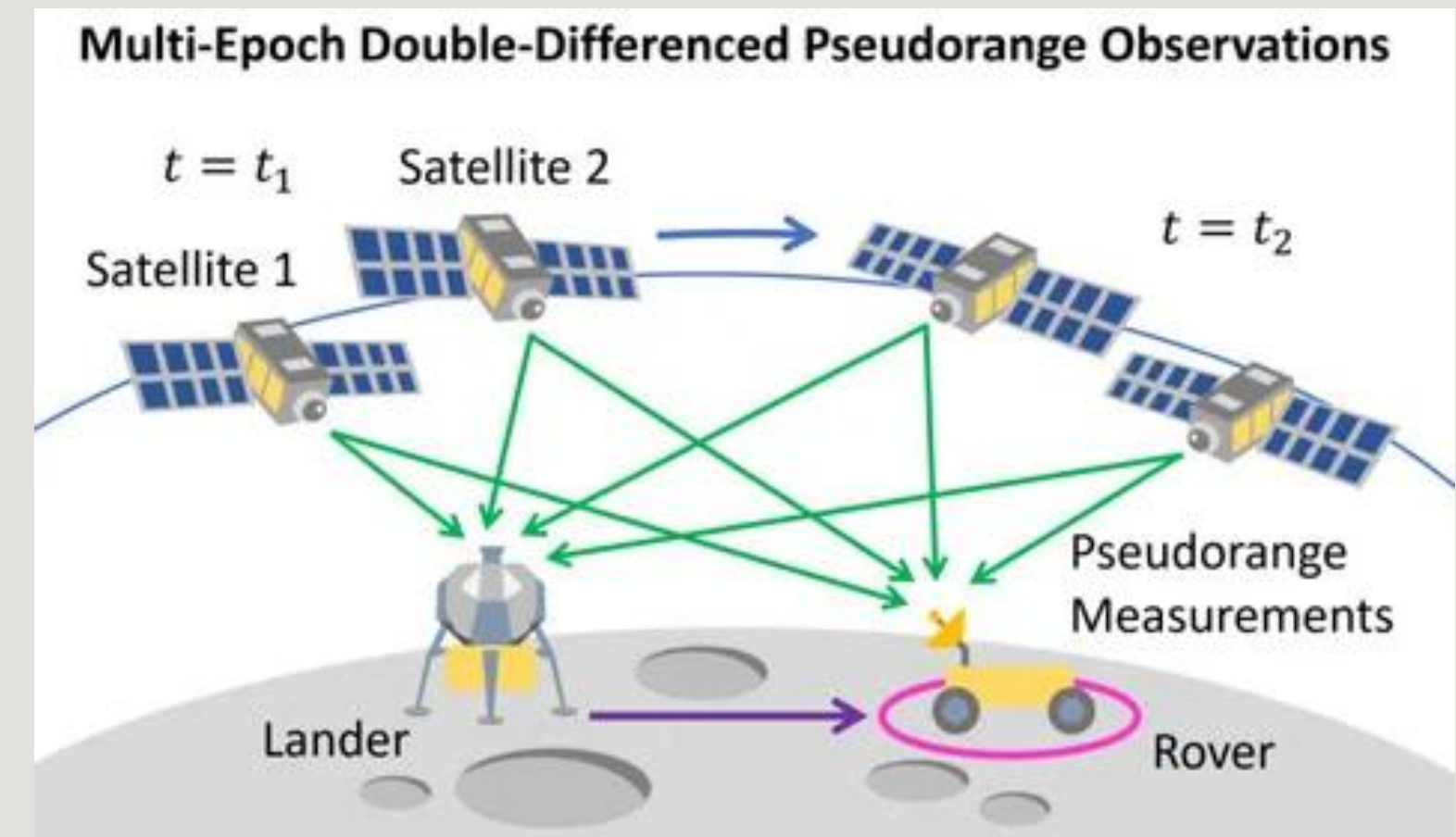
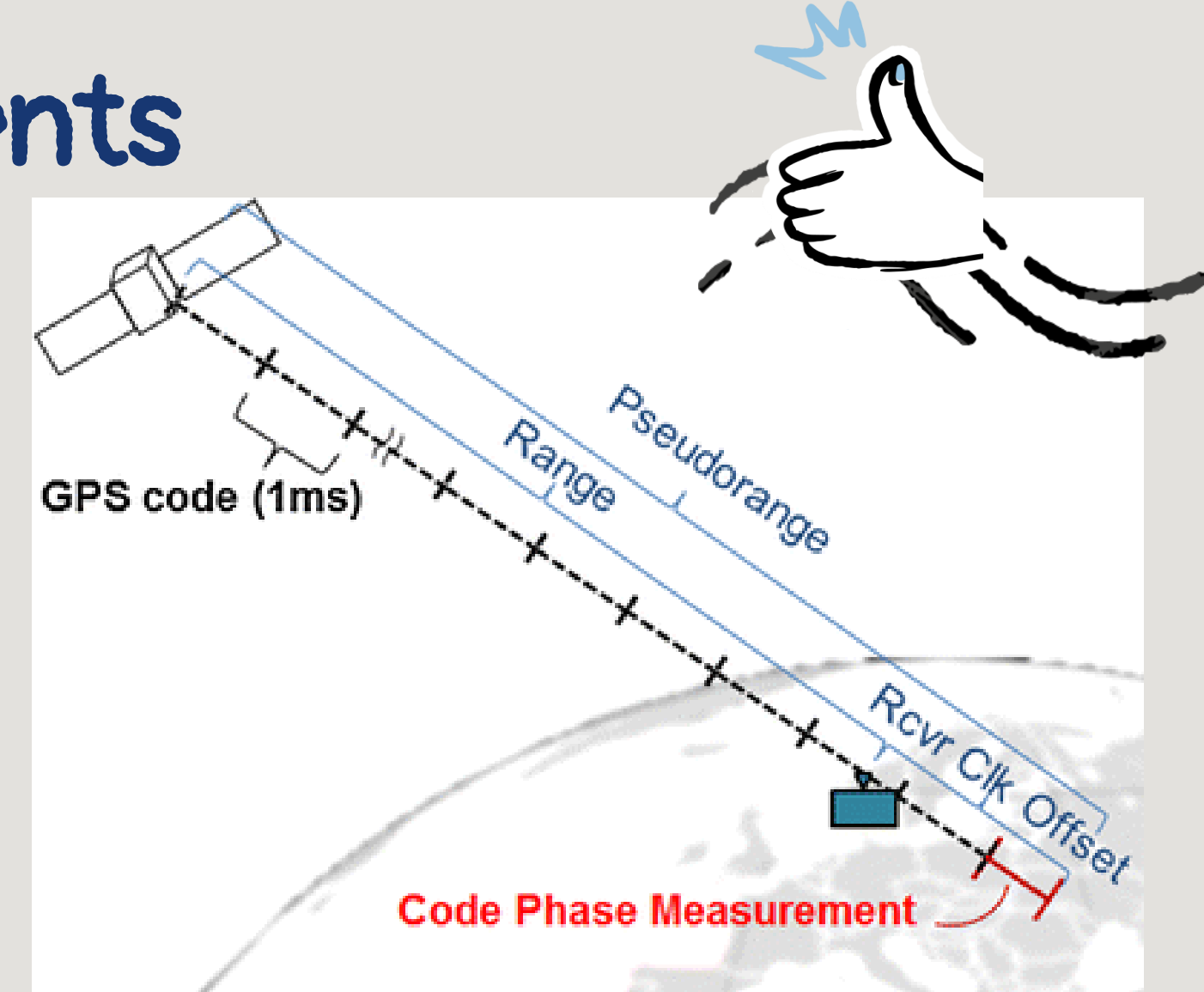
## 05 Code-Based Positioning (Pseudorange Method) (Used for satellite identification & ranging)

Each satellite transmits a unique PRN code to distinguish itself.

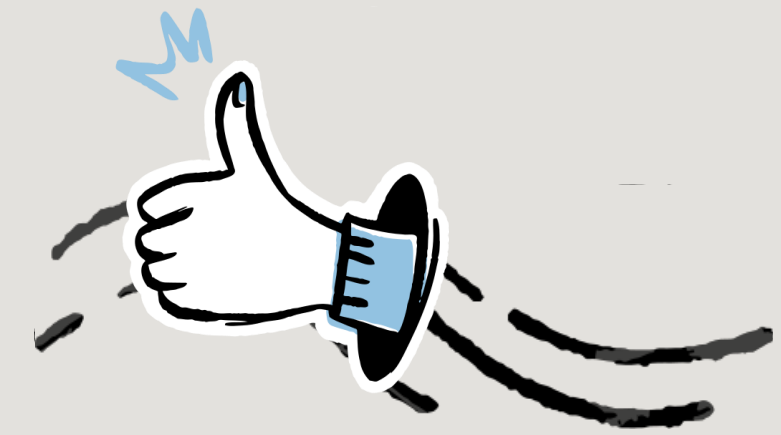
- The receiver compares the received and generated PRN codes to measure the time delay.

Types:

- C/A Code – Civilian use (L1 frequency).
- P(Y) Code – Encrypted military use.
- L2C, L5, E1, E5, B1, B2 Codes – Modernized signals for improved accuracy.

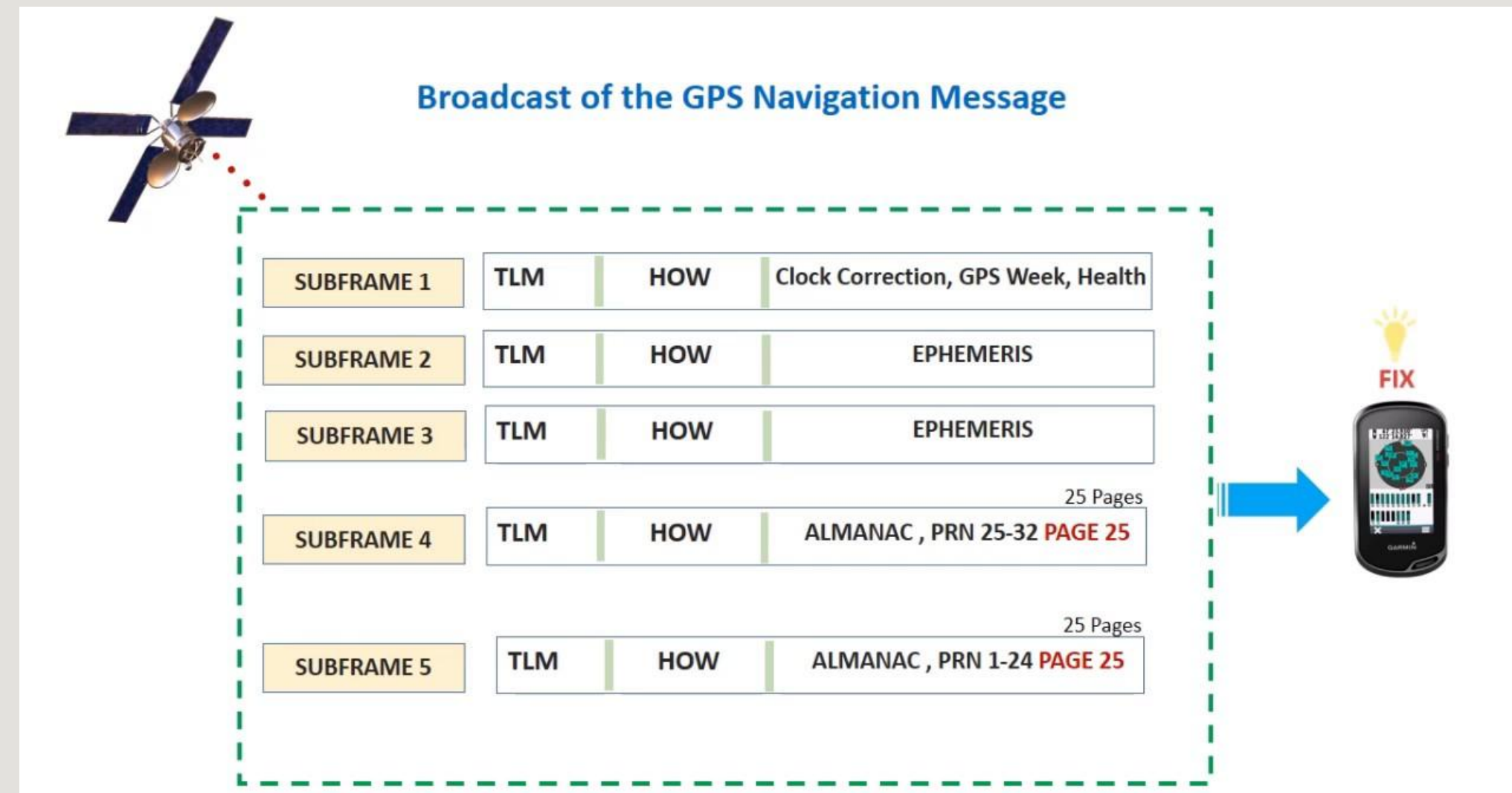


# GNSS Signal Components

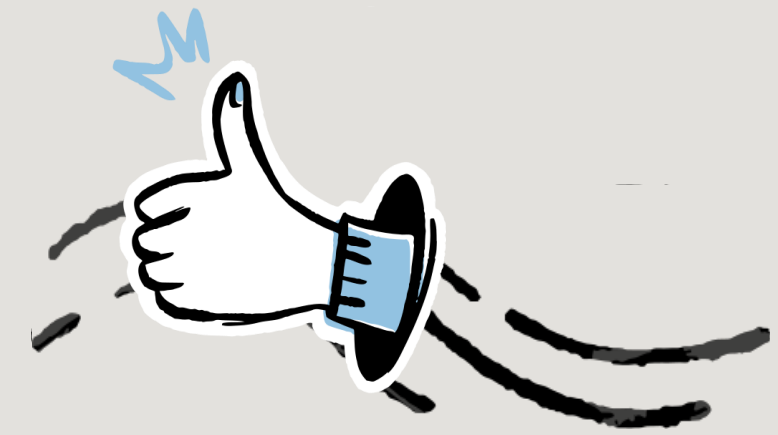


## 06 Navigation Message (contains essential satellite data)

- **Ephemeris Data** – Provides precise satellite position.
- **Clock Correction Data** – Corrects satellite clock errors.
- **Ionospheric & Tropospheric Corrections** – Reduces atmospheric signal delays.
- **Almanac Data** – Approximate positions of all satellites for quick acquisition.



# Sources of GNSS Errors



## Satellite-Related Errors

01

- **Orbit Errors** – Small deviations in satellite orbits affect positioning.
- **Clock Errors** – Tiny inaccuracies in satellite clocks cause signal timing errors.

## Signal Propagation Errors

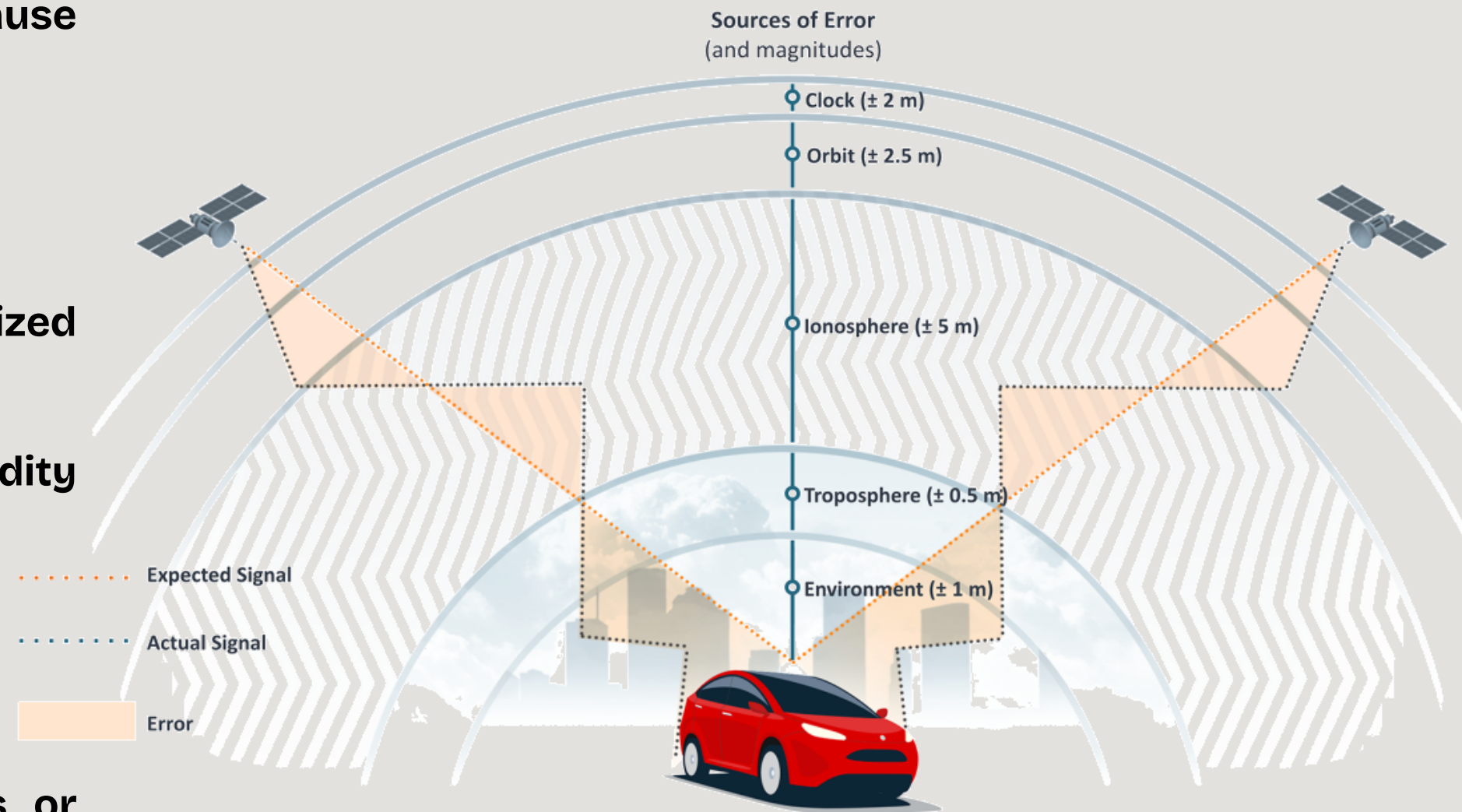
02

- **Ionospheric Delay** – GNSS signals slow down due to ionized particles in the upper atmosphere.
- **Tropospheric Delay** – Signal bending caused by humidity and temperature changes.

## Receiver-Related Errors

03

- **Multipath Effect** – Signals bounce off buildings, trees, or terrain before reaching the receiver, causing errors.
- **Receiver Noise** – Internal errors in the GNSS receiver introduce minor inaccuracies.



# GNSS Fundamental Frequencies – Derived from Base Clock

**Base Clock Rate ( $f_0$ ) = 10.23 MHz**

- **All GPS L-band** frequencies are integer multiples (or offsets) of this clock.
- Keeps synchronization between satellite & receiver.
- Other GNSS use similar reference frequencies (slightly different base rates).
- **Galileo uses 10.23 MHz equivalent for E1, E5**
- GLONASS uses FDMA, so base frequency differs slightly
- **NavIC L5 is same as GPS L5 (1176.45 MHz)**

| Signal    | Formula          | Frequency (MHz)              |
|-----------|------------------|------------------------------|
| L1        | $154 \times f_0$ | $154 \times 10.23 = 1575.42$ |
| L2        | $120 \times f_0$ | $120 \times 10.23 = 1227.60$ |
| L5        | $115 \times f_0$ | $115 \times 10.23 = 1176.45$ |
| C/A code  | $f_0 \div 10$    | 1.023 MHz                    |
| P(Y) code | $f_0$            | 10.23 MHz                    |

- **International GNSS Day** is officially celebrated on **October 23** each year.
- The date 10/23 was chosen because it mirrors the foundational GPS frequency of 10.23 MHz, which serves as the cornerstone of L-band signals for over 100 navigation satellites today.

# GNSS Observation Types



## Code measurements → metre-level

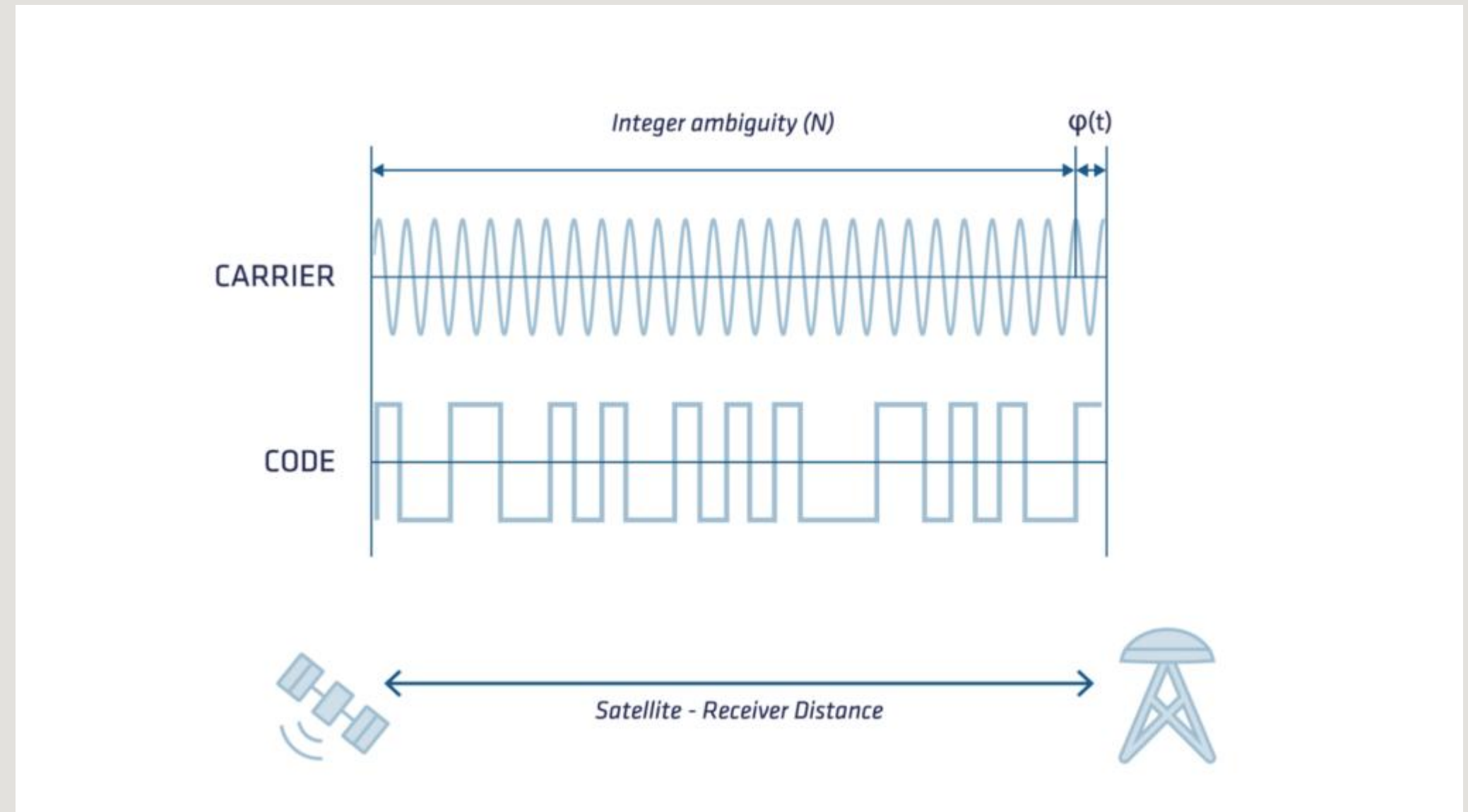
01

- Uses PRN codes like C/A, P(Y)Code chips are wide → metre-level accuracy
- Easy to acquire and track
- Main use: navigation, SBAS, rough positioning

## Carrier-phase → mm-level

02

- Uses the continuous sinusoidal carrier (L1, L2, etc.)
- Wavelength is small ( $\approx 19\text{--}24\text{ cm}$ ) → mm-cm accuracy
- Very precise but harder to track
- Main use: RTK, static, geodetic applications

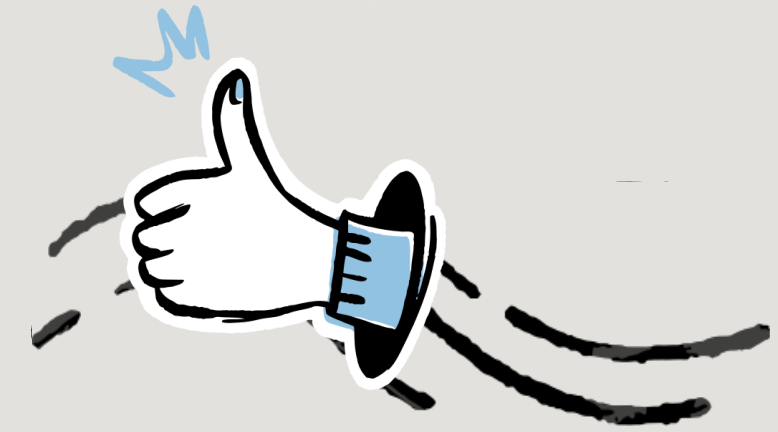


# Differential Techniques

## Single Differencing (SD)

01

- Difference between two receivers tracking the same satellite
- Removes satellite clock error and reduces common biases
- Produces a receiver-to-receiver relative observation



## Double Differencing (DD)

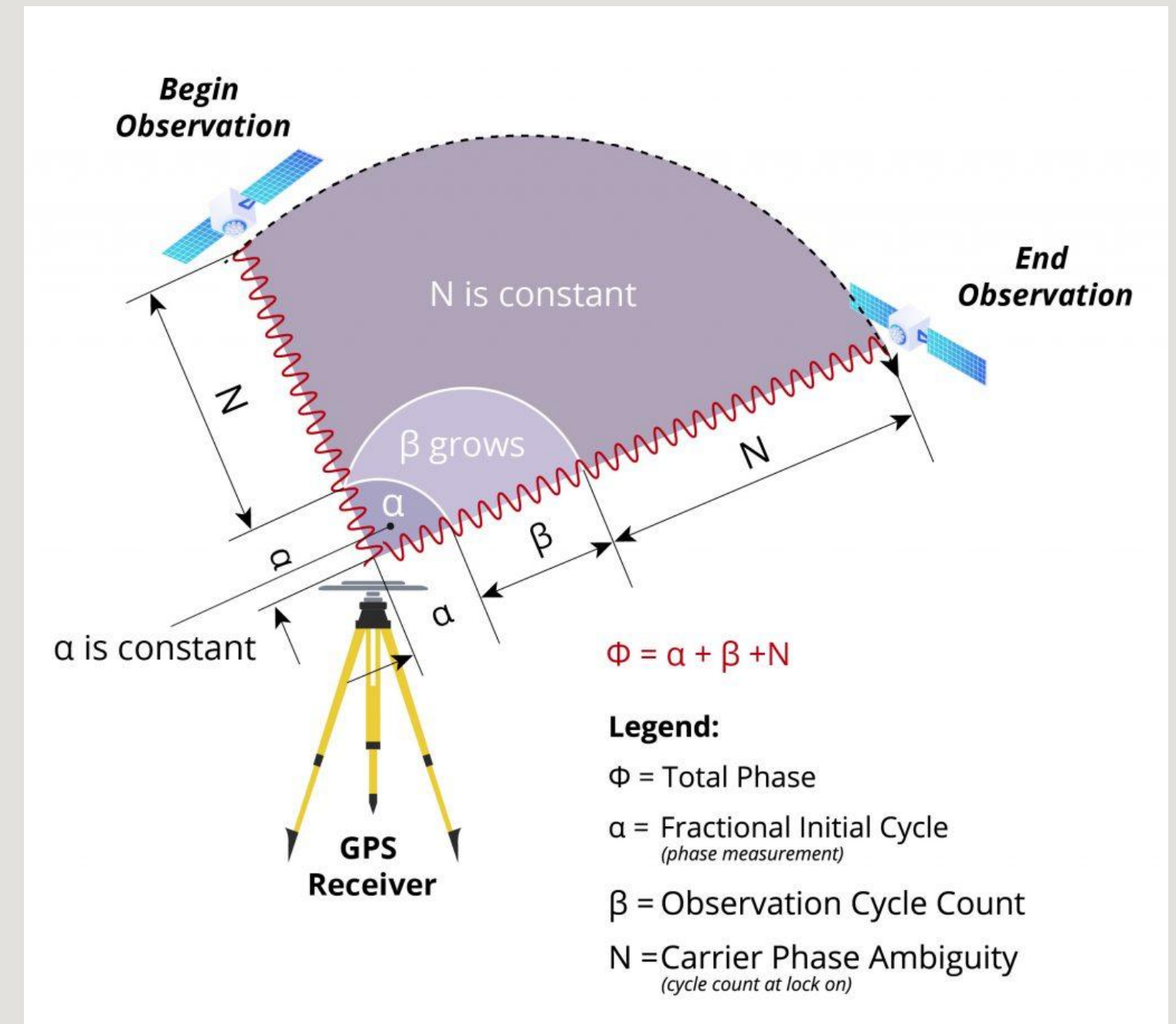
02

- Difference of two single differences using two receivers and two satellites
- Eliminates both satellite and receiver clock terms
- Provides the fundamental observable for precise baseline computation

## Integer ambiguity

03

- At start, receiver knows fractional phase but not how many full carrier cycles are between satellite and receiver
- This unknown whole-number count = integer ambiguity
- Correctly resolving it → fixed solution (cm accuracy)
- If unresolved → float solution (dm accuracy)

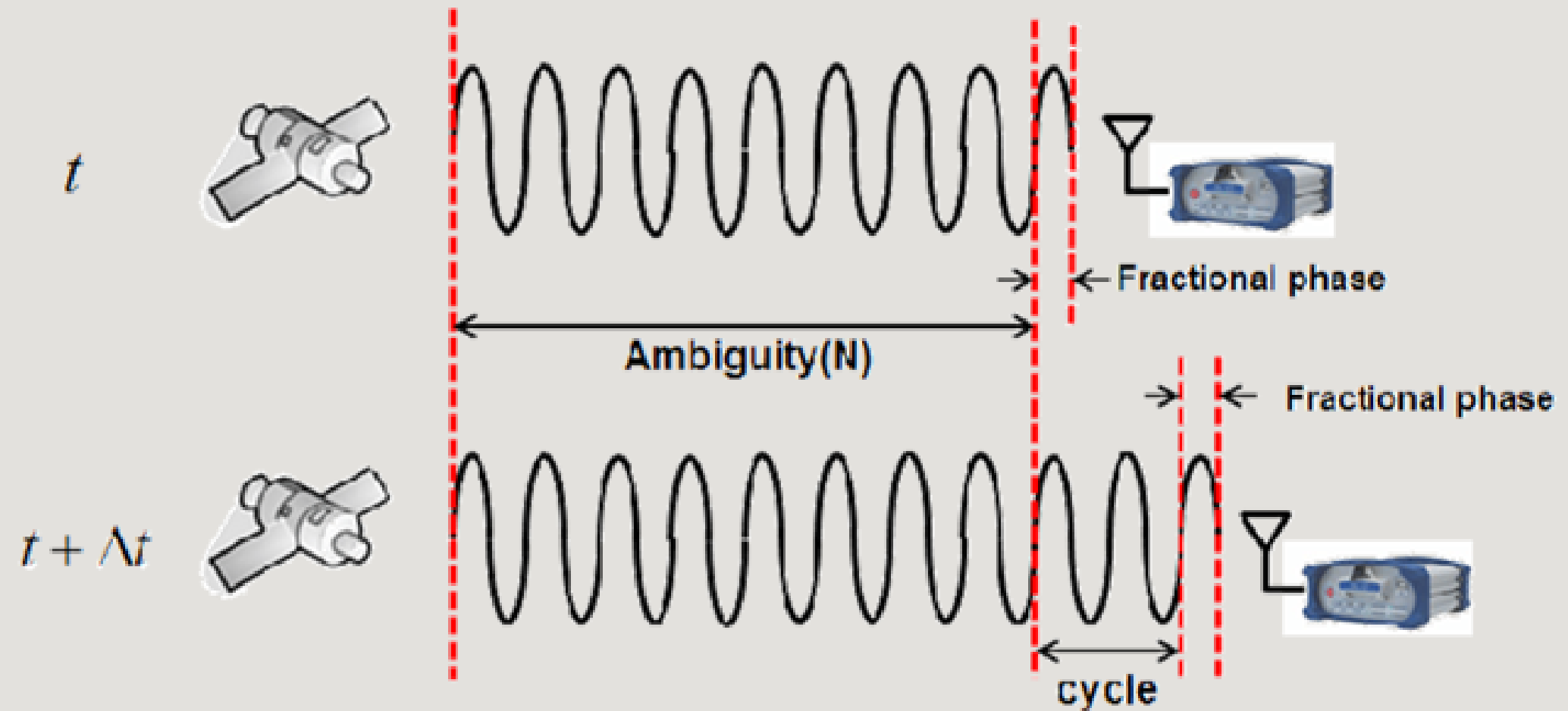


# Integer Ambiguity

- Imagine the GNSS satellite sends a **wave** to your receiver, made up of tiny repeating cycles.
- Your receiver can measure the **last incomplete bit** of the wave, but it doesn't know **how many full waves** passed before that.
- That count of **whole waves** is called the *Integer Ambiguity*.

## Why it matters in surveying:

- If we figure out this count correctly, we can get **centimeter-level accuracy**.
- If we get it wrong, the position will be off.



# How We Solve Integer Ambiguity

## 1. Take Multiple Opinions

- Your receiver listens to signals from many satellites at once.
- Like asking several friends the same question — the common answer is usually correct.

## 2. Use Two Measuring Scales

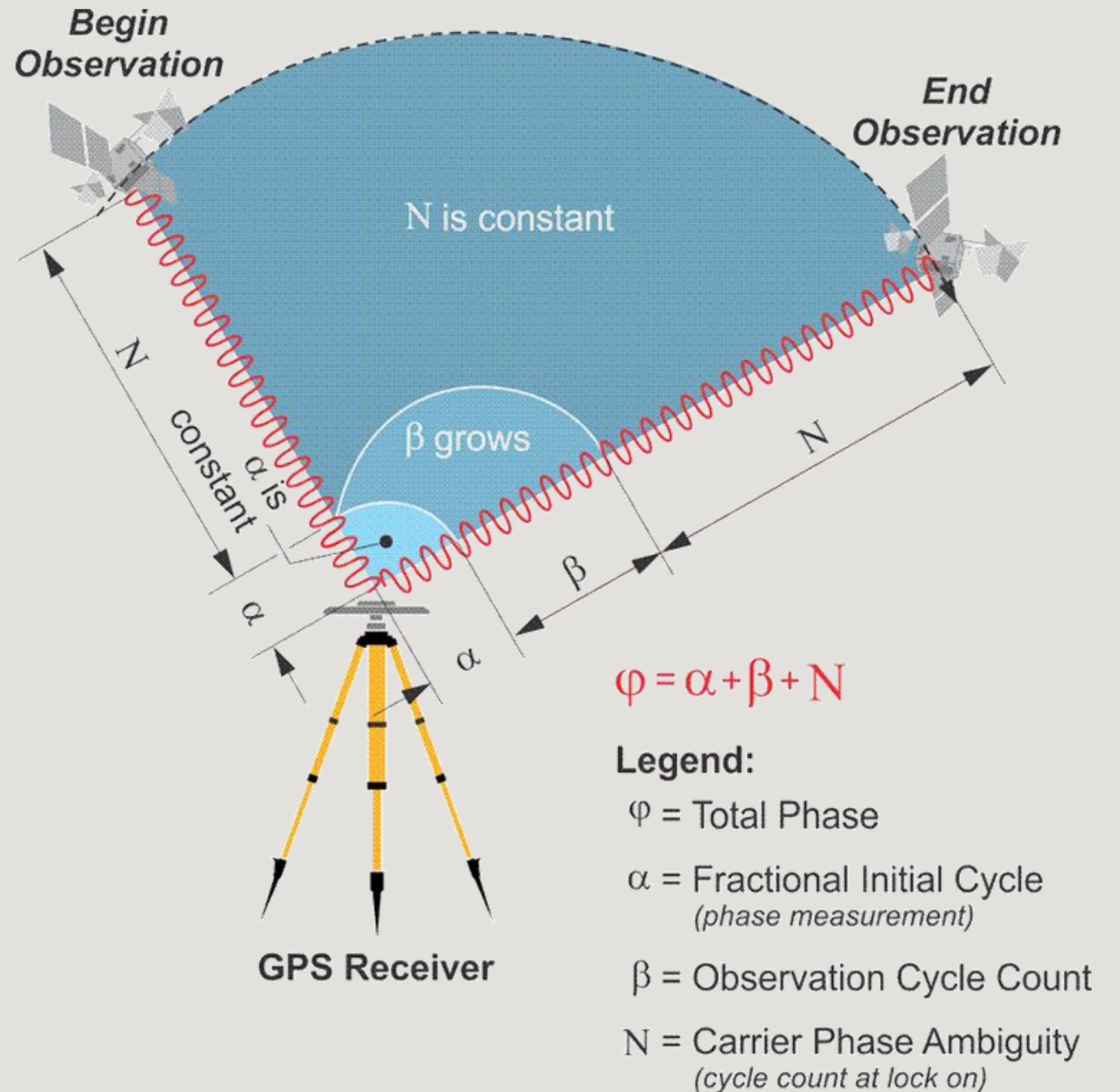
- Dual-frequency GNSS works like having two rulers with different markings.
- Comparing both helps remove “atmosphere fog” so the count becomes clearer.

## 3. Check Over Time

- The receiver keeps tracking as you stay still or move.
- With more time, it spots a clear pattern and locks the exact number.

## 4. Lock It In

- Once the number of whole waves is certain, the system “fixes” it.
- From then on, your position jumps from meter-level to centimeter-level accuracy.



# Types of Geometric Factors

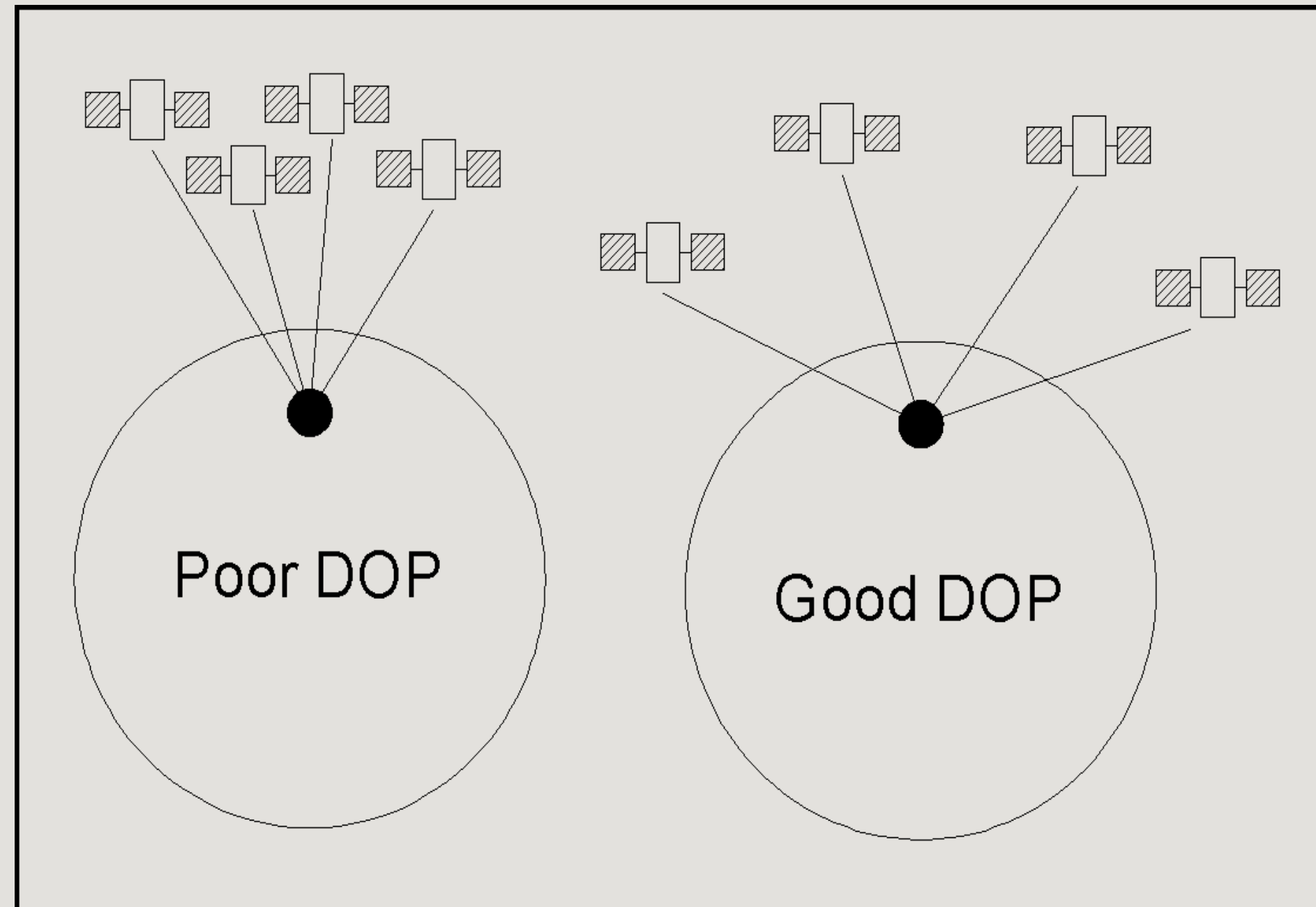
The strength of figure of the satellites is taken into consideration by the dilution of precision (DOP) factor

- Depends on number of satellites.
- Depends on location of satellites.

**Good receivers determine which satellites will give the lowest GDOP**

Different kinds of DOPs

- HDOP (horizontal)
- VDOP (vertical)
- PDOP (position) (3-D component)
- TDOP (time)
- GDOP (geometric) (PDOP and TDOP)



# GNSS Positioning Techniques

01

## Static GNSS

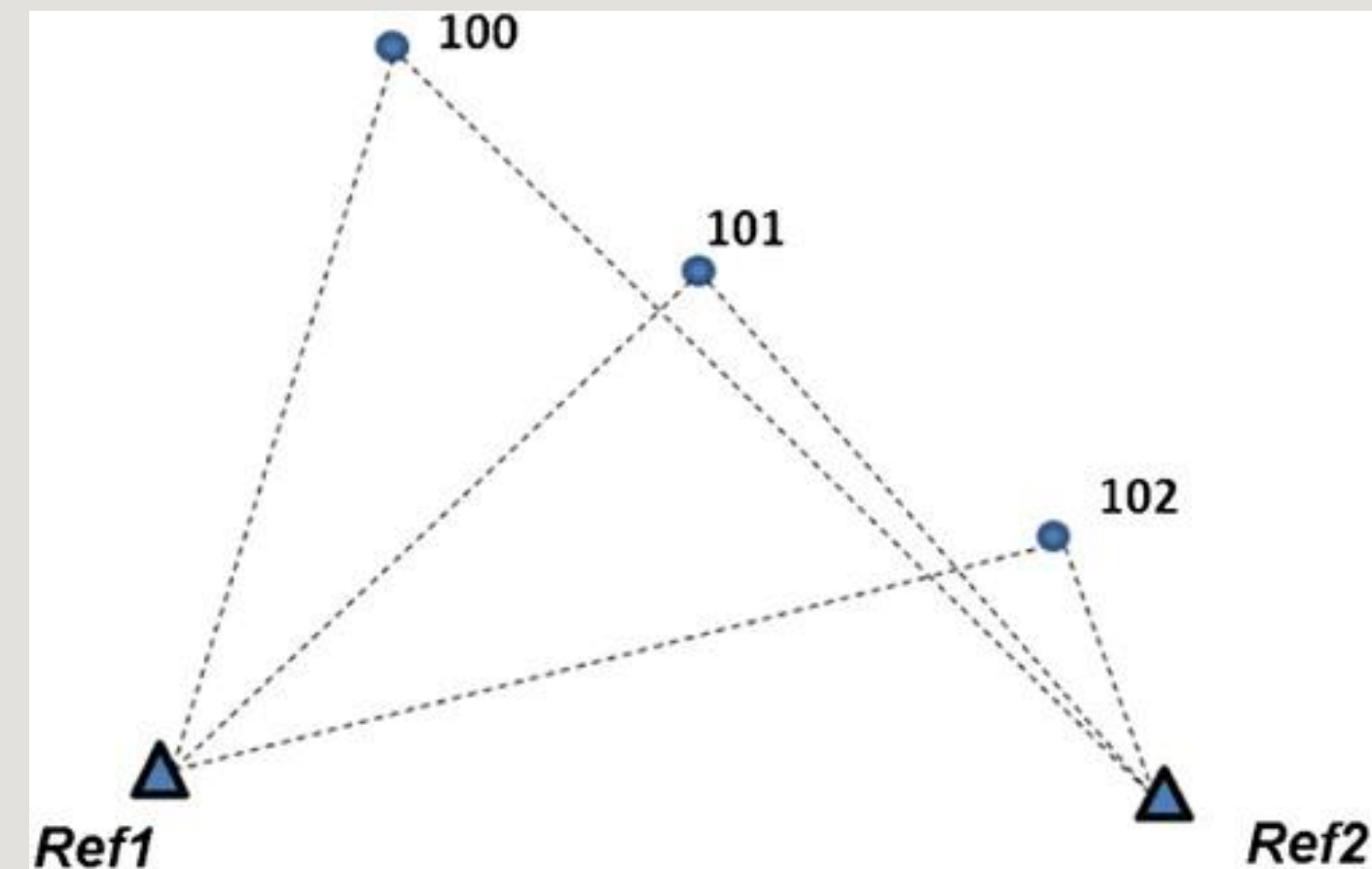
- **Method:** Two or more receivers remain stationary for long observation periods (several hours).
- **Accuracy:** Millimeter-level ( $\sim 1 - 5$  mm) after post-processing.
- **Setup:** Requires a base station or a known reference point.
- **Use Case:** Geodetic control, tectonic studies, deformation monitoring.



02

## Rapid Static GNSS

- **Method:** Similar to Static GNSS, but with shorter observation times (15 min to 1 hour).
- **Accuracy:** Centimeter-level ( $\sim 1$  cm) after post-processing.
- **Setup:** Requires a base station or a CORS network.
- **Use Case:** Quick control point surveys where RTK is unavailable.



# GNSS Positioning Techniques



03

## Real-Time Kinematic (RTK)

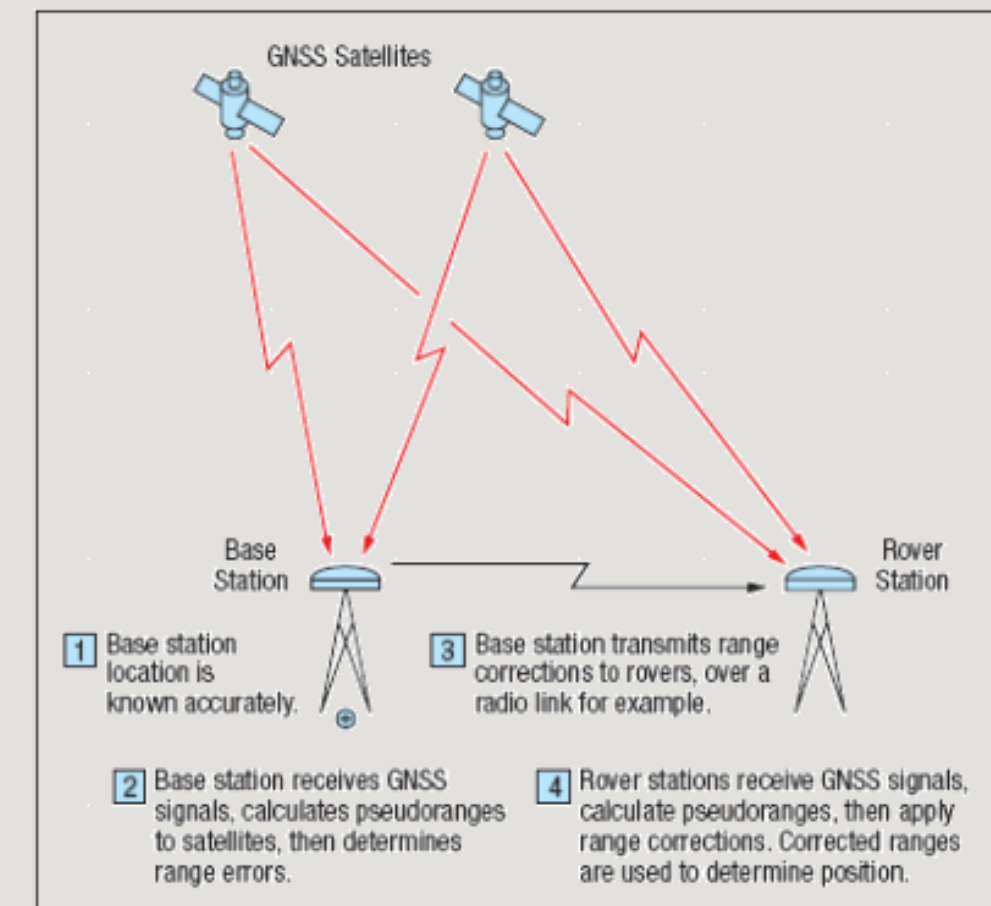
- **Method:** Uses a base station and a rover, with real-time corrections sent via radio, internet, or satellite.
- **Accuracy:** Centimeter-level (~ 1 – 2 cm) in real-time.
- **Setup:** Requires continuous communication between the base and rover.
- **Use Case:** Precision surveying, agriculture, drone mapping.



04

## Differential GNSS (DGNSS)

- **Method:** Base station transmits corrections to a rover, improving standard GNSS accuracy.
- **Accuracy:** Sub-meter (~ 0.5 – 1 m).
- **Setup:** Requires a base station but does not need carrier-phase measurements like RTK.
- **Use Case:** GIS mapping, marine navigation, vehicle tracking.

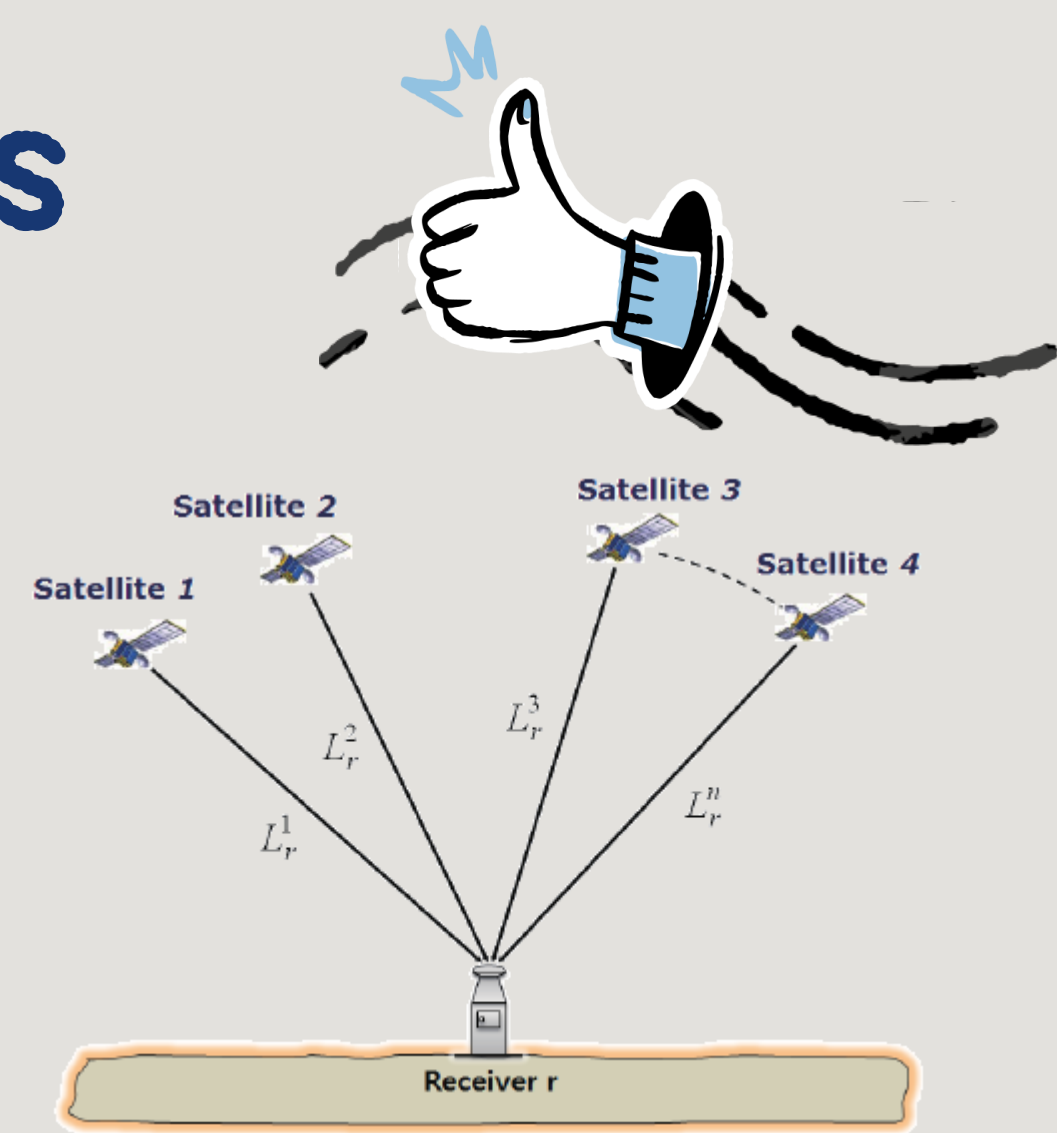


# GNSS Positioning Techniques

05

## Precise Point Positioning (PPP)

- **Method:** Uses a single GNSS receiver, relying on precise satellite clock/orbit corrections.
- **Accuracy:** Decimeter to centimeter (~5–10 cm) after post-processing or real-time PPP.
- **Setup:** No base station required, but global corrections must be available.
- **Use Case:** Remote areas where local base stations are unavailable.



06

## Network RTK (NRTK)

- **Method:** Uses a network of continuously operating reference stations (CORS) instead of a single base station.
- **Accuracy:** Centimeter-level (~1–2 cm) in real-time.
- **Setup:** Rover connects to CORS via the internet to receive corrections.
- **Use Case:** Urban areas, precision agriculture and surveying over large regions.
- **Example Services:** VRS (Virtual Reference Station), MAC (Master-Auxiliary Concept), FKP (Flächen-Korrektur Parameter).

# Overview of CORS



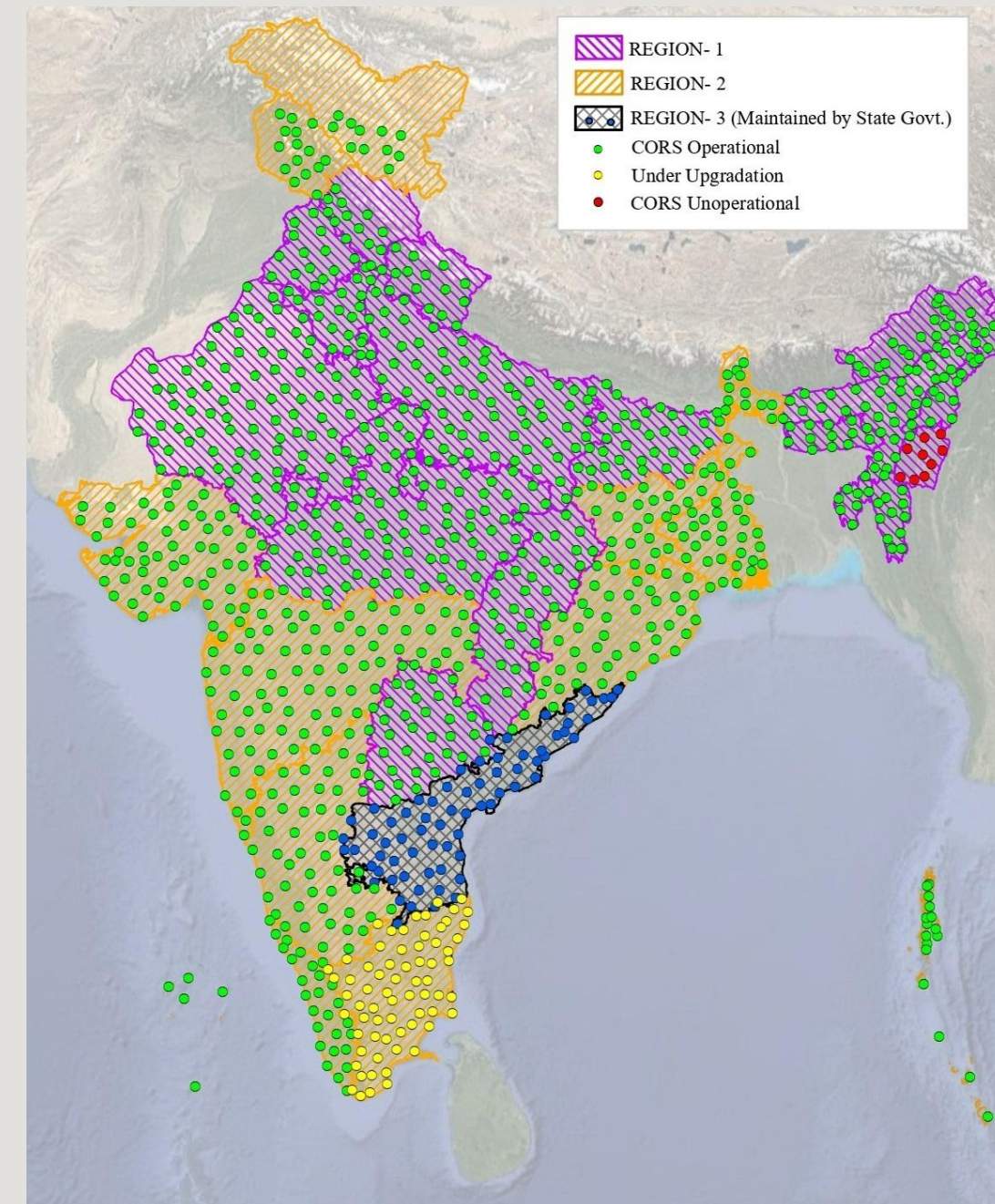
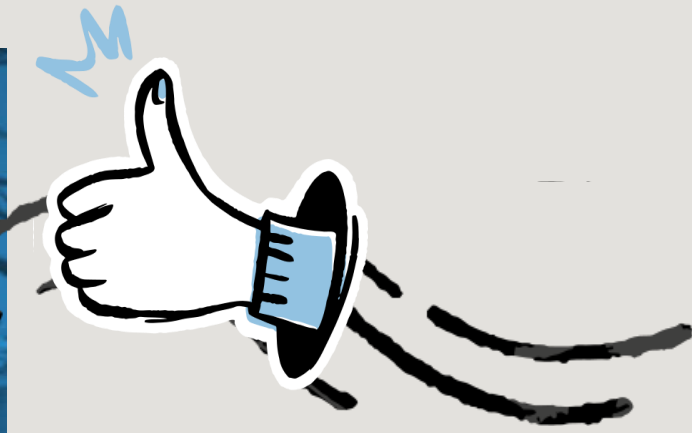
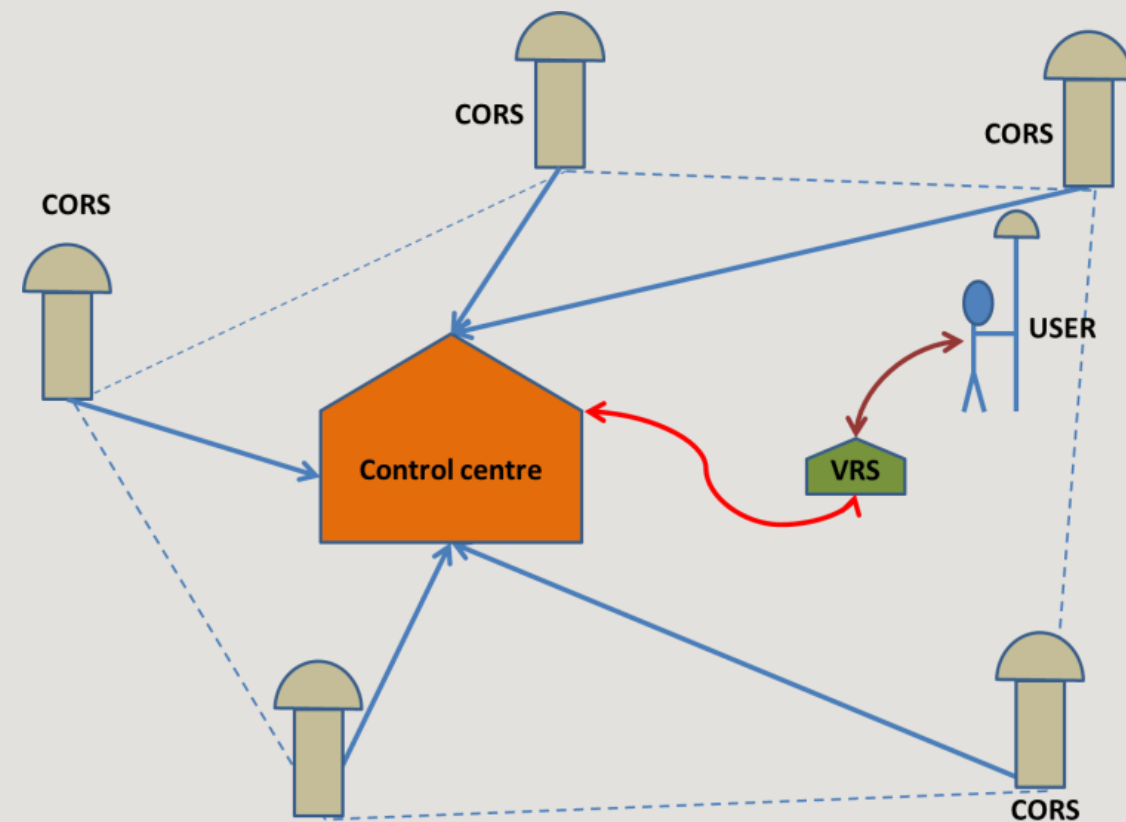
**NSRF:** A reference system for locating positions on the Earth's surface. It defines the coordinate system that enables precise mapping and navigation.

**India's Horizontal Datum:** India transitioned from the Everest Spheroid to the globally accepted ITRF (International Terrestrial Reference Frame), aligning with modern global geodetic standards.

**CORS (Continuously Operating Reference Stations):** Permanent GNSS stations providing real-time, high-precision positioning data. CORS network supports applications like Real-Time Kinematic (RTK) positioning, enhancing geospatial accuracy across various sectors.

# Concept and Working Principle of NRTK

- **CORS Network** continuously collects GNSS data.
- The Network Control Center (NCC) at Survey of India, Dehradun processes data and generates correction models.
- The correction data is sent to users (rovers) through mobile internet (GPRS/4G/NTRIP).
- The rover applies the corrections and computes centimeter-level accuracy positions.



# Components of NRTK System



- **Reference Stations (CORS Network)**

- A network of Continuously Operating Reference Stations (CORS) equipped with high-precision GNSS receivers.
- Collects raw satellite data and sends it to the control center.
- Ensures stability and long-term accuracy for positioning.
- Covers large geographic areas to improve correction accuracy.

- **Control Center and Correction Services**

- Processes GNSS data from multiple CORS stations.
- Computes correction models (e.g., VRS, FKP, MAC techniques).
- Distributes real-time corrections to users via NTRIP (Networked Transport of RTCM via Internet Protocol).
- Maintains quality control and system health monitoring.



- **Rover Receivers and Their Role in Field Surveys**

- Mobile GNSS units (rovers) used for high-precision field measurements.
- Connect to the CORS network via the internet (4G/5G/NTRIP client).
- Apply correction data from the control center to compute cm-level accurate positions.
- Used in urban mapping, surveying, and precision applications.



# Data Correction Techniques in NRTK

NRTK uses advanced error modeling techniques to improve positioning accuracy:

## Virtual Reference Station (VRS)

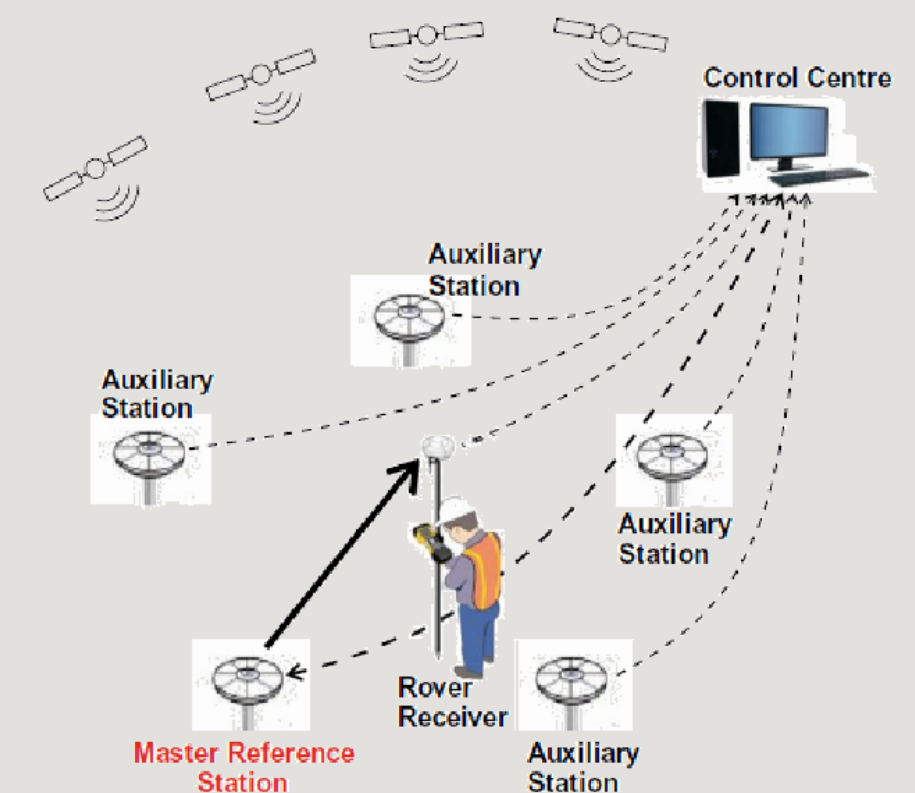
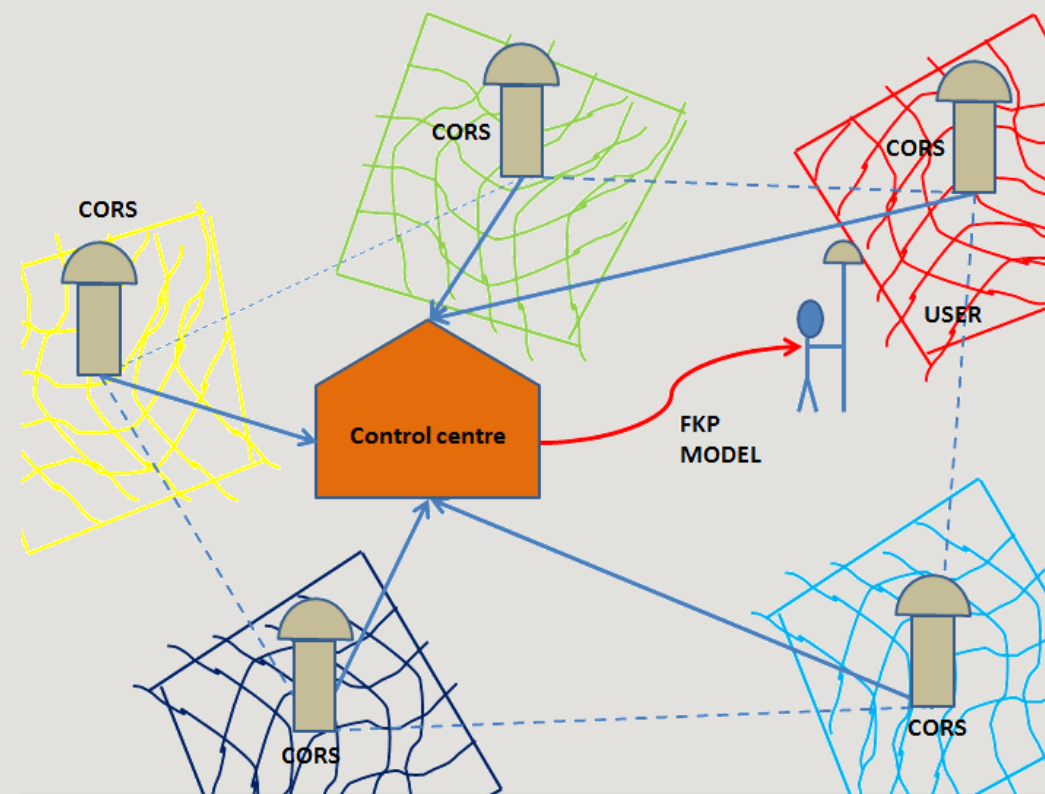
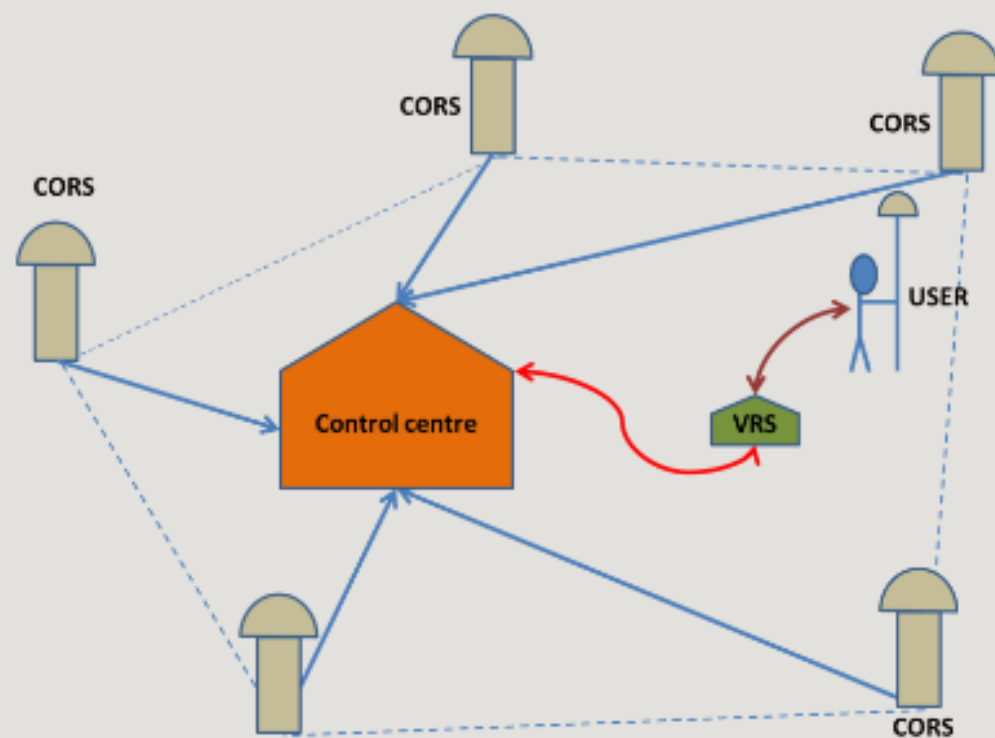
- Creates a virtual base station near the rover using data from multiple CORS stations.
- Provides consistent accuracy by interpolating corrections.
- Works well for single-user applications.

## Flächen-Korrektur-Parameter (FKP)

- Uses a mathematical grid model to estimate correction parameters over a region.
- Provides fast and reliable corrections but may have lower accuracy in highly variable atmospheric conditions.

## Master-Auxiliary Concept (MAC)

- Transmits raw GNSS data from a master station and additional data from auxiliary stations.
- Allows the rover to compute its own corrections dynamically.
- More efficient than VRS, supports multiple users simultaneously.



# Advantages of CORS Over Conventional GNSS Techniques

## Key Advantages

### 1. Cost and Labor Efficiency

1. Eliminates the need to set up a base reference station for each user.
2. Reduces operational costs and labor.

### 2. Enhanced Accuracy and Consistency

1. Homogeneous and consistent accuracy through centralized processing.
2. Uses the same functional and stochastic modeling, and datum.

### 3. Wider Coverage with Fewer Stations

1. Larger area coverage with fewer reference stations compared to single reference RTK.

### 4. Reliability and Redundancy

1. Higher reliability and availability of RTK corrections.
2. Improved redundancy ensures continued operation even if one station fails.

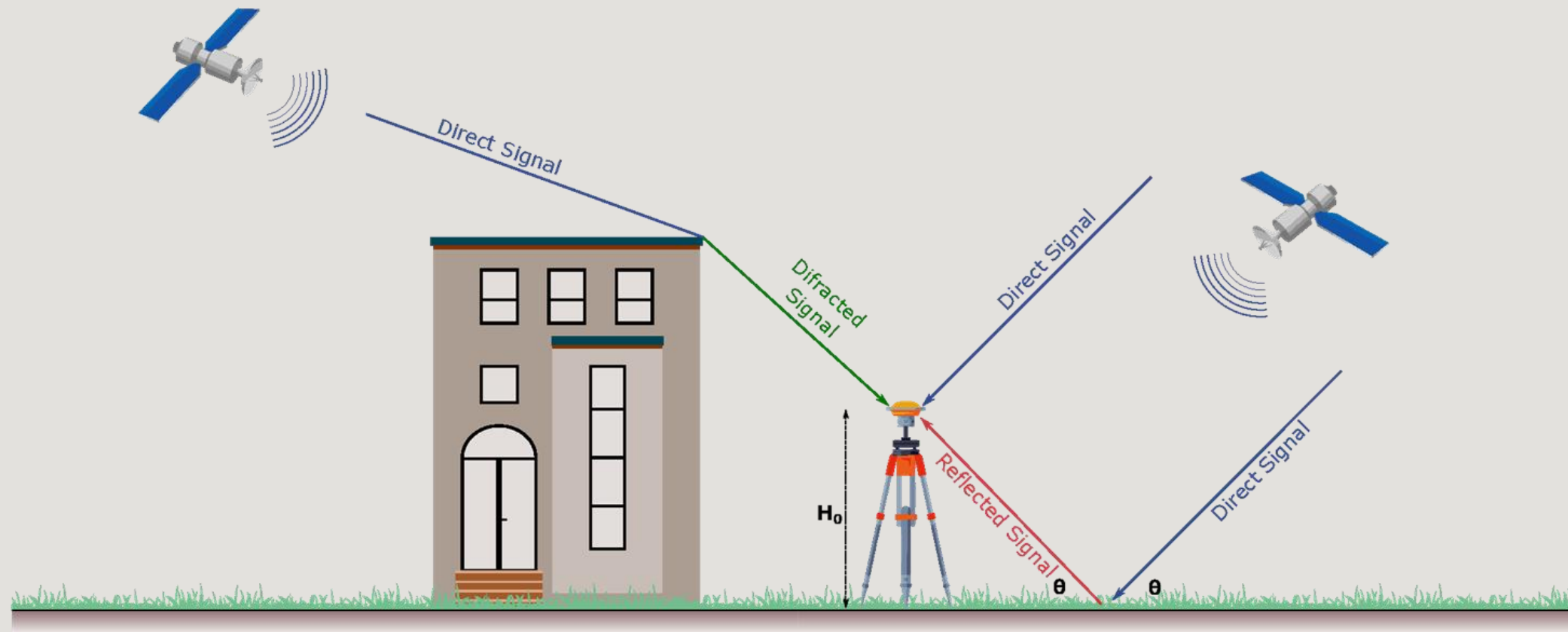
### 5. Support for Multiple Users and Applications

1. Capable of supporting multiple users simultaneously.
2. Used by state and central agencies for surveying, mapping, and development activities.



# Antenna Specifications and Error Mitigation

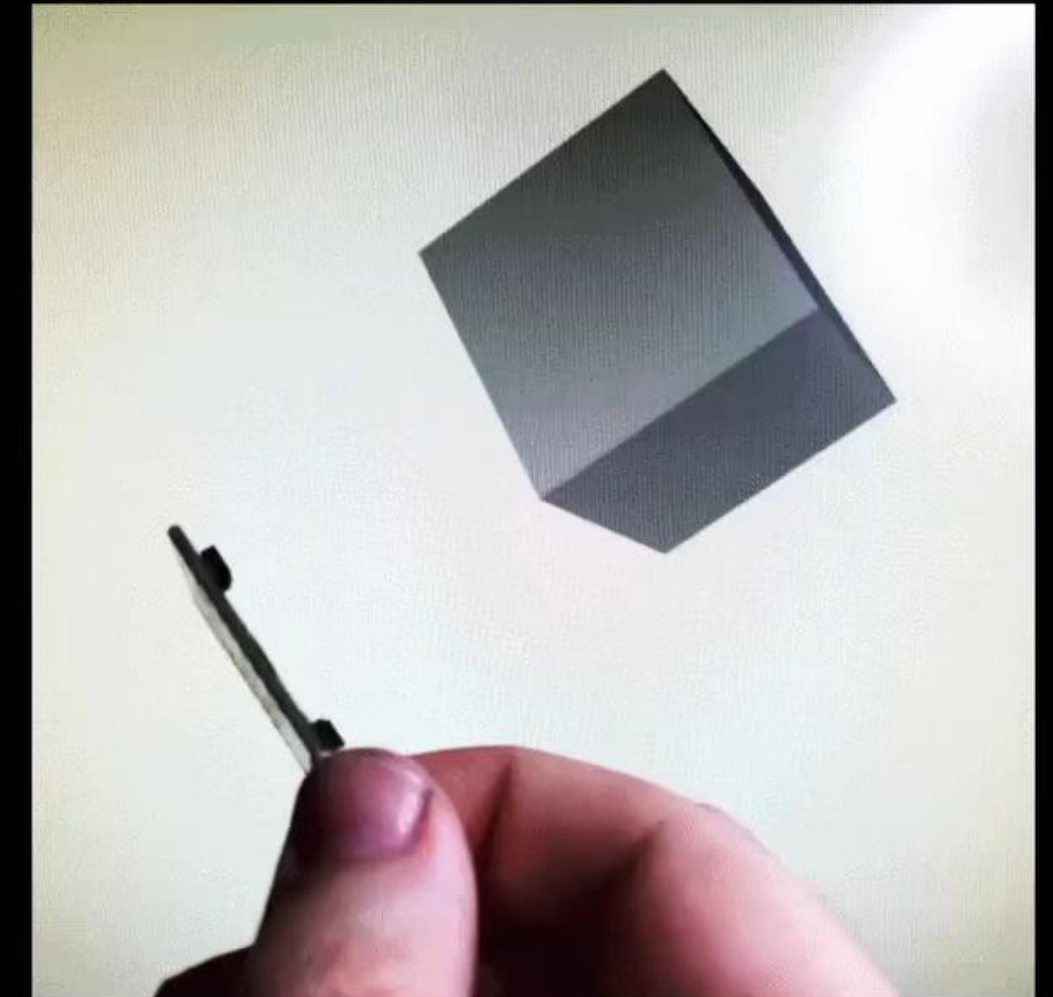
- **Geodetic-Grade Antenna** – High-precision choke ring or helical antenna recommended.
- **Multipath Rejection** – Must have built-in multipath mitigation technology to reduce signal interference.
- **Phase Center Stability** – Low phase center variation for consistent accuracy.
- **Low Noise Amplifier (LNA)** – Enhances weak GNSS signals in challenging environments.
- **Weatherproof & Rugged Design** – IP67-rated enclosure for field operations.



# Functionality of IMU

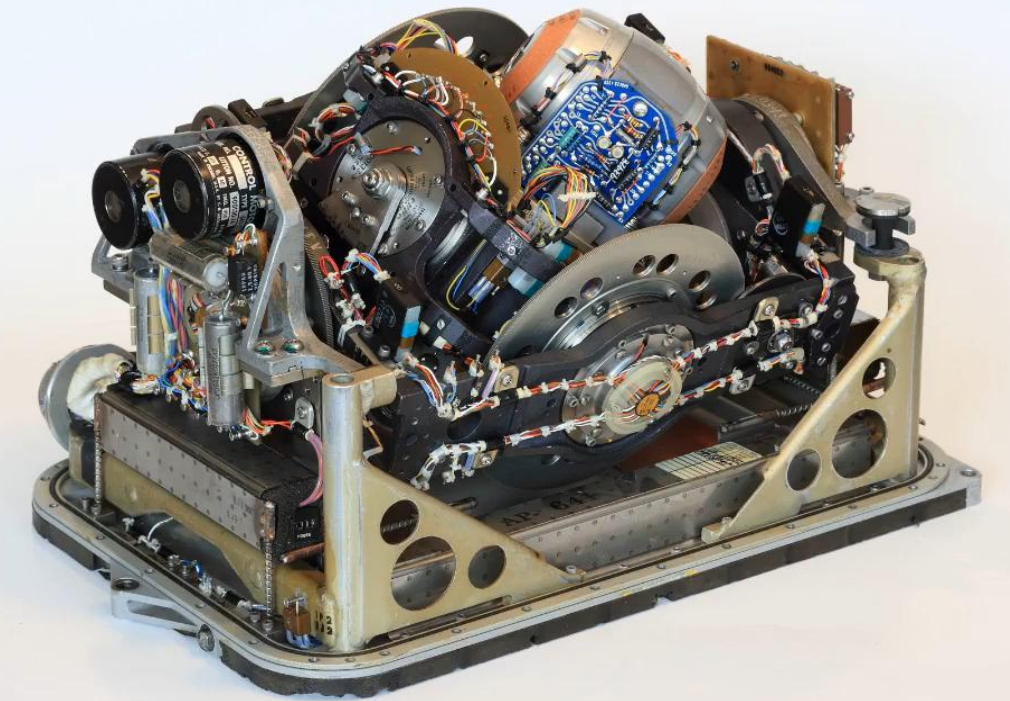
- **Detecting Changes in Velocity and Orientation:**
  - **Velocity Changes:** Monitored by accelerometers detecting linear acceleration.
  - **Orientation Changes:** Monitored by gyroscopes detecting rotational rates.
  - **Magnetometers** assist in stabilizing orientation by providing absolute directional reference.
- **Importance in Navigation and Stability:**
  - **Navigation:** Ensures accurate tracking and control in vehicles, drones and other systems.
  - **Stability:** Helps maintain stability and correct deviations in dynamic environments.

**INERTIAL SENSORS are composed of accelerometers and gyroscopes.**



# What is an INS?

- Self-contained navigation system that calculates position, orientation, and velocity.
- 
- $\text{INS} = \text{IMU} + \text{Positioning Algorithm}$
- It Integrates accelerometer and gyroscope data with GNSS inputs.
- Components:
  - Accelerometers: Measure linear acceleration.
  - Gyroscopes: Measure rotational movement.
  - Algorithms: Integrate sensor data to compute position and velocity.
- Key Features:
  - Independence: Operates without external signals.
  - Continuous Updates: Provides real-time data on position and motion.
  - Drift: Accumulates errors over time, often corrected with GNSS.



# Challenges in Canopy Areas in Urban/Forest Environments

## GNSS Signal Obstruction Due to Trees and Buildings

### **Dense Foliage & Tall Buildings Block Signals**

- GNSS signals struggle to penetrate thick tree canopies and urban structures.
- Leads to frequent loss of satellite lock and poor satellite geometry (low GDOP).

### **Urban Canyons**

- High-rise buildings create narrow sky visibility, reducing the number of available satellites.
- Causes delayed signal reception, affecting real-time positioning.

## Multipath Errors and Their Effects on Accuracy

### **Signal Reflection & Interference**

- Signals bounce off surfaces like glass buildings, metal structures, and water before reaching the receiver.
- Causes incorrect distance measurements, leading to position shifts and accuracy degradation.

### **Loss of RTK/NRTK Fix**

- Multipath and obstructions increase the time required for fixing ambiguities.
- May result in floating solutions or complete positioning failures.

Thank you

